

Quintile Return Book

Average monthly return of each factor's quintile portfolios and the long / short quintiles selected from them. 269 factors, ordered by in-sample L–S CAGR.

HOW TO READ

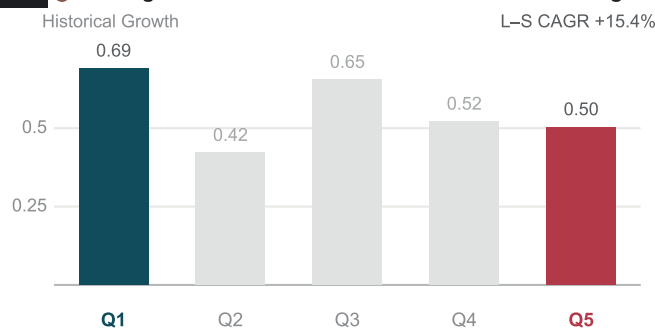
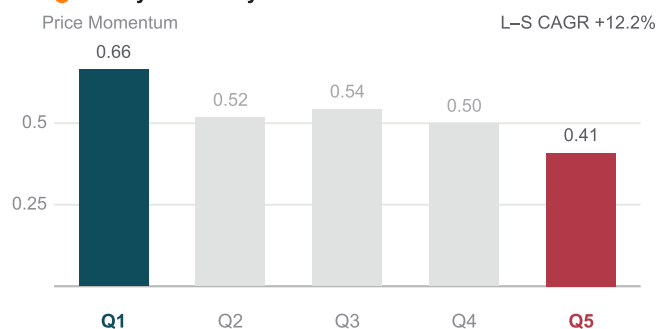
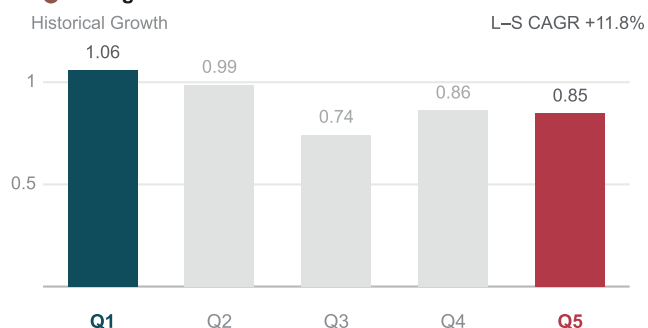
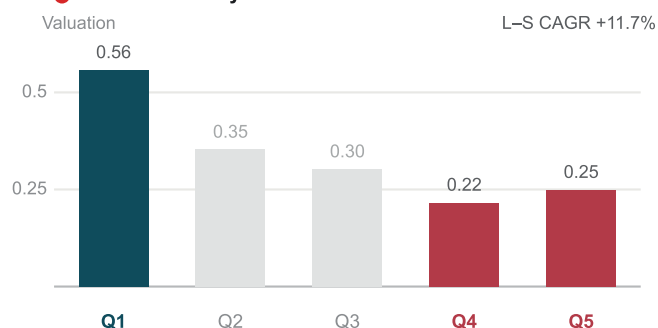
■ Long ■ Short ■ Neutral

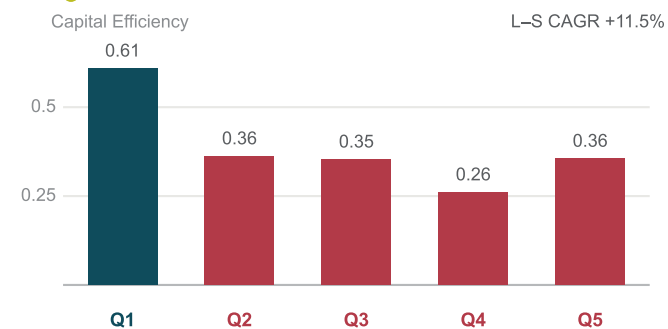
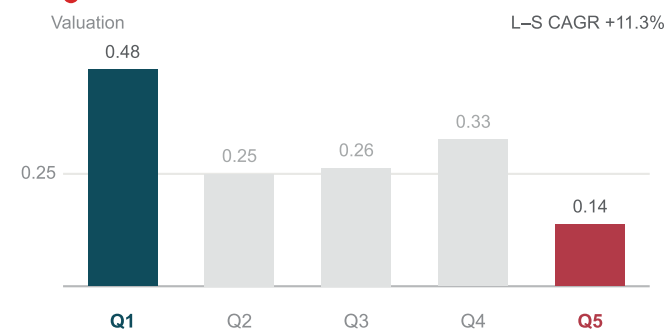
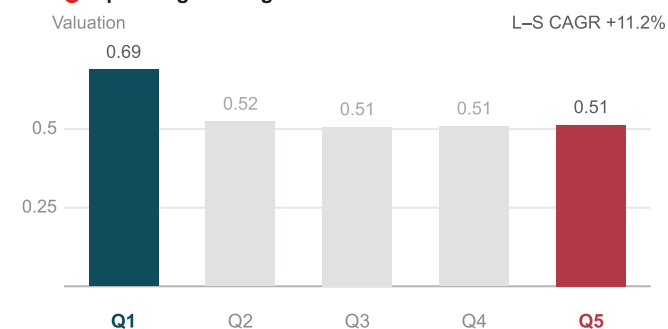
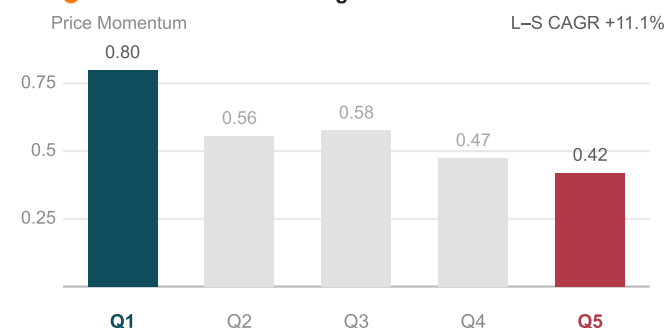
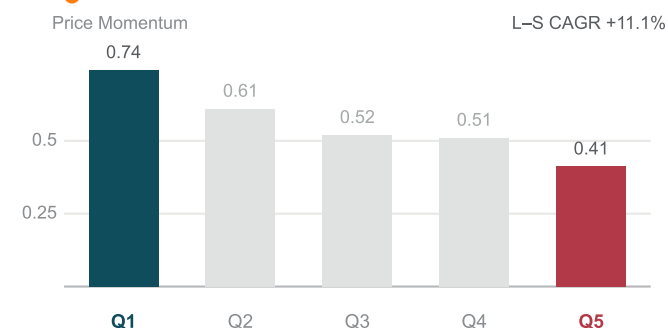
Quintiles are selected as long or short by performance against a spread-based threshold — not fixed to Q1 / Q5. Grey quintiles are left out of the portfolio.

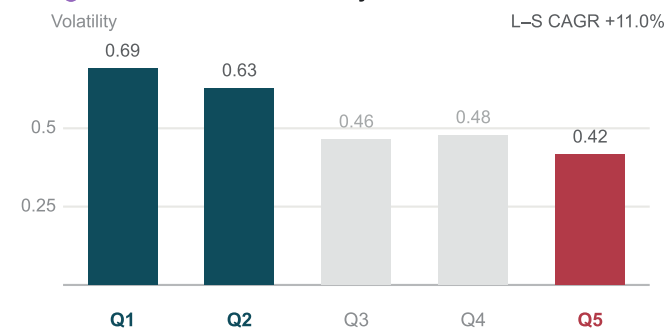
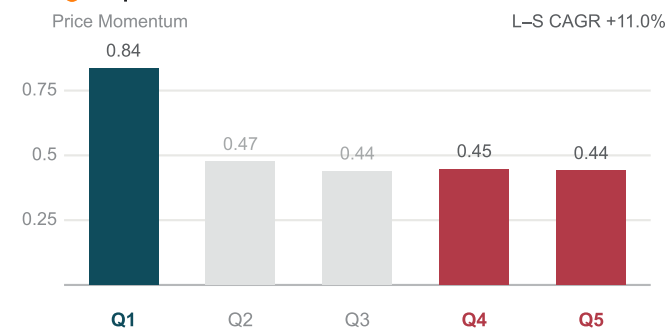
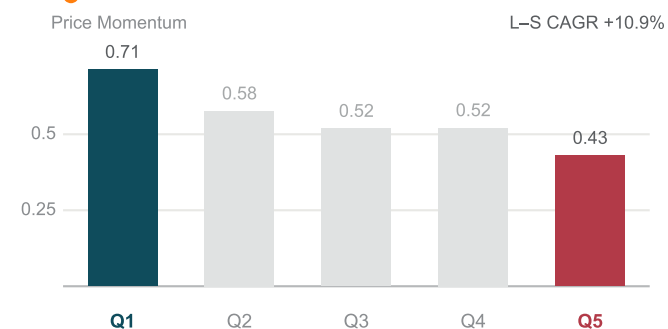
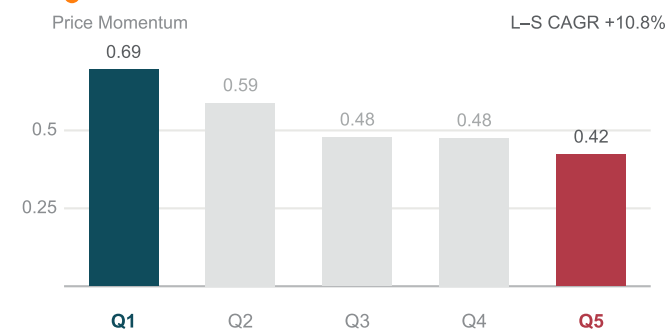
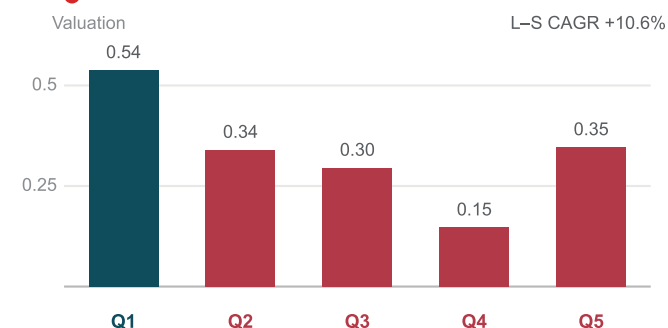
FACTOR STYLES

● Valuation	41	● Price Momentum	55
● Earnings Quality	55	● Size	1
● Analyst Expectations	24	● Historical Growth	55
● Capital Efficiency	28	● Volatility	10

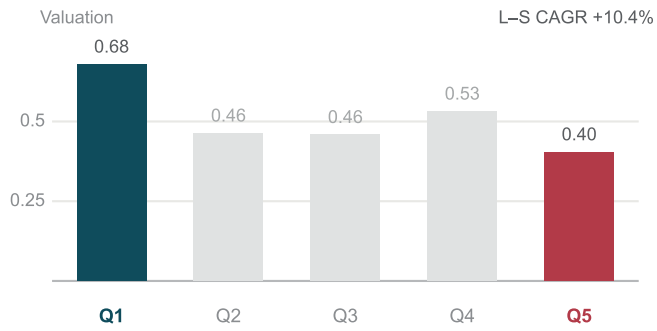
001 In current model portfolio (50 factors)

001 ● Num of Qtrs of Pos Chgs in TTM Cash Flows/TTM Sales**002 ● Trailing 12-Month Receivables and Inventories/Trailing 12-M...****003 ● Number of Consecutive Qtrs of Pos. Changes in TTM Cash Flows****004 ● 50 Day to 200 Day Stock Price****005 ● Change of Asset Turnover****006 ● Current Liability to Price**

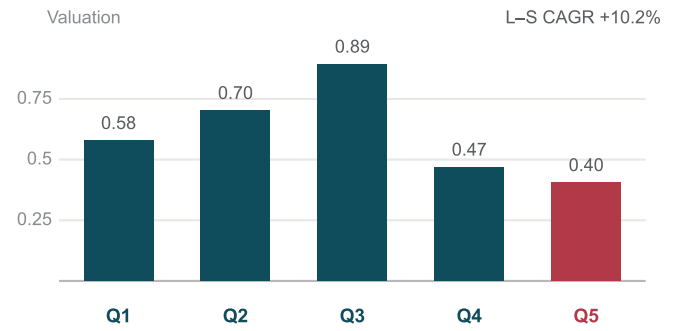
007 ● **Tobin's Q Ratio**008 ● **39W Lagged Return**009 ● **Pretax Income Per Share to Price**010 ● **Operating Earnings to Price Ratio**011 ● **26W Relative Price Strength**012 ● **9 Month Price Momentum**

013 ● **12M Realized Price Volatility**014 ● **Slope of 52W Price Trend**015 ● **15/36 Week Price Ratio**016 ● **4W to 52W Price Oscillator**017 ● **Analyst Dispersion for FY2 EPS**018 ● **Book to Price**

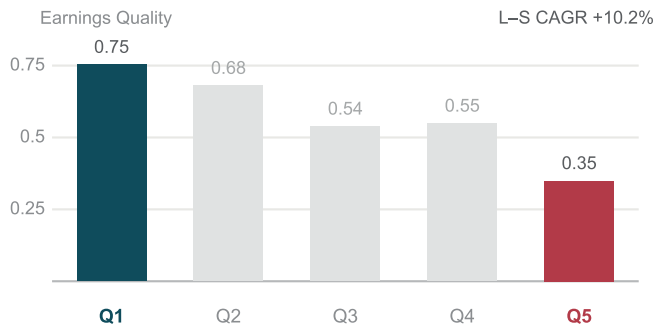
019 ● Normalized EP



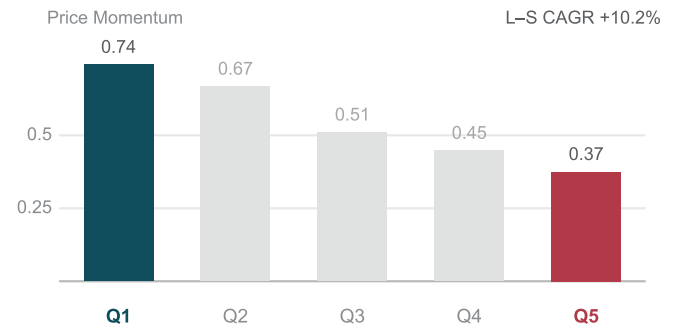
020 ● Indicated Annual Dividends to Price Ratio



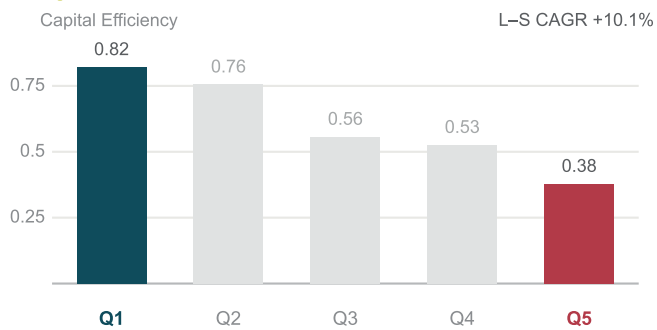
021 ● Operating Earnings to Assets Ratio



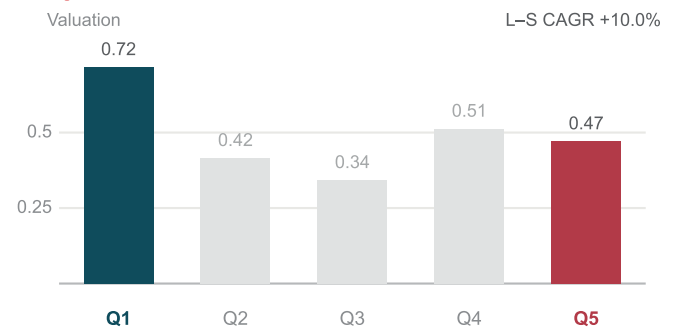
022 ● Percent Above 260-Day Low

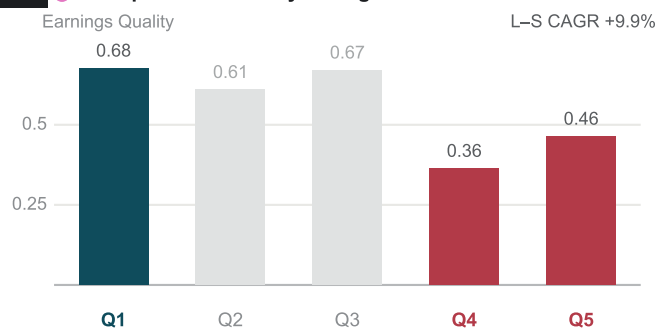
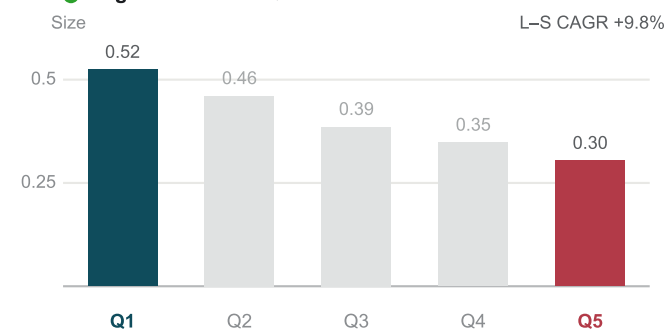
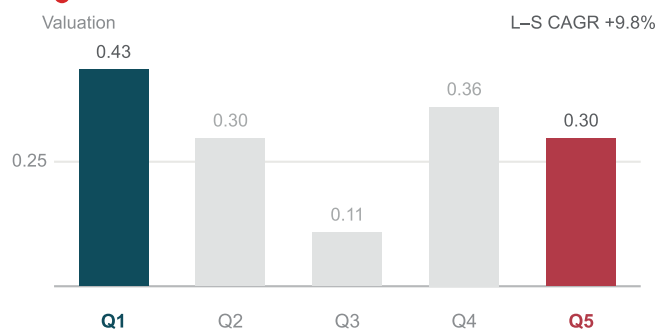
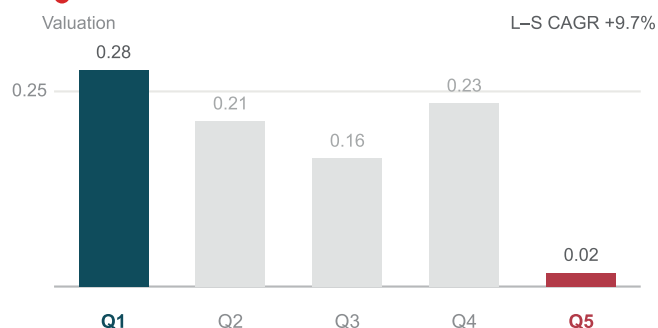


023 ● Return on Assets

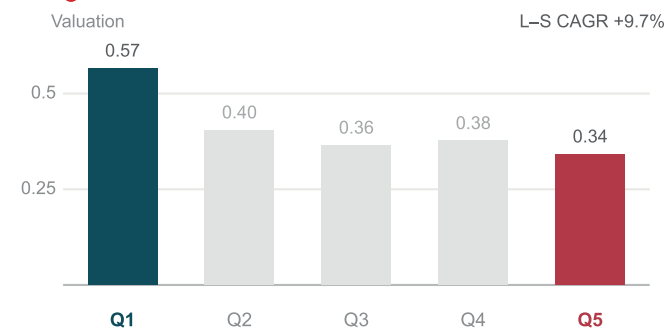


024 ● EBITDA to Price

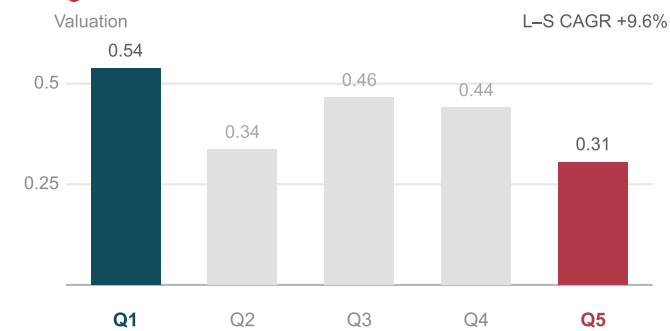


025 ● **Stability of Change in Number of Holders**026 ● **Unexpected Inventory Change**027 ● **Log of Total Last Quarter Assets**028 ● **Assets to Price Ratio**029 ● **6M Chg in Target Price**030 ● **Cash to Price**

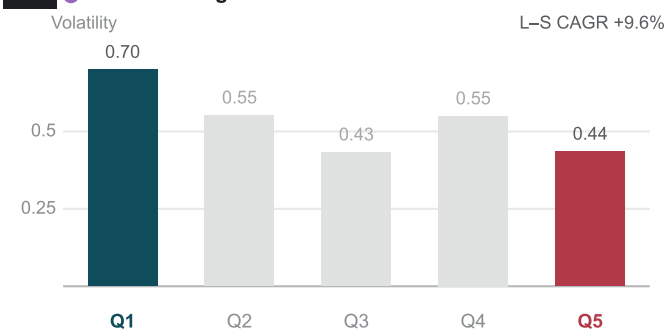
031 ● Sales to Price Ratio



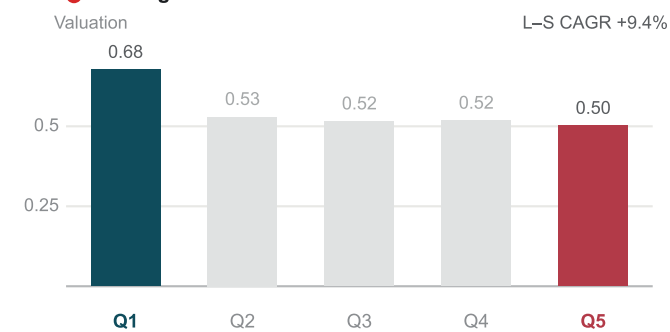
032 ● Growth Flow to Price



033 ● Book Leverage



034 ● Earnings to Price



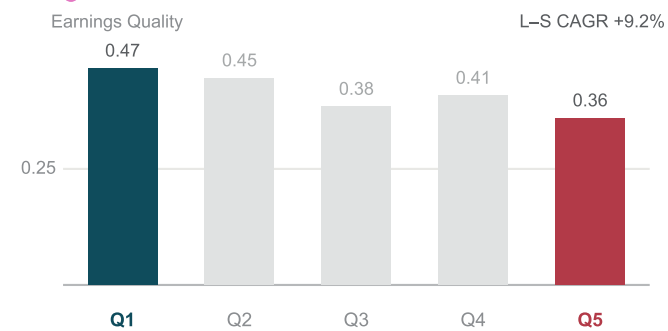
035 ● Cash Flow to Total Asset



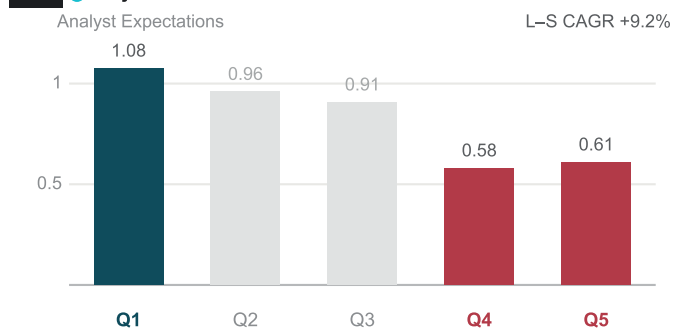
036 ● Number of Consecutive Qtrs of Positive Changes in TTM Quart...



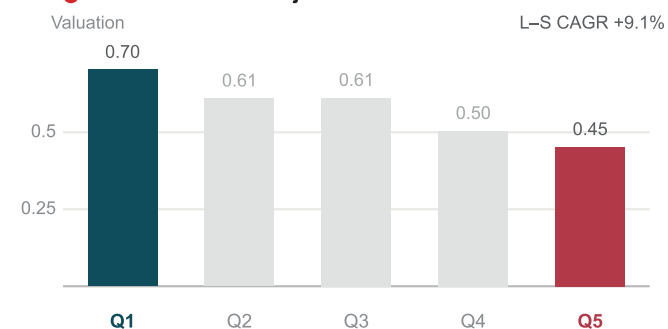
037 ● ROA 20 Qtr Standard Deviation



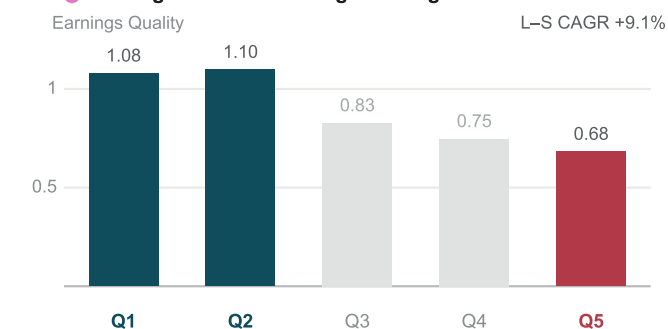
038 ● Adjusted 3M Revision in FY1 EPS Estimate



039 ● Inverse P/E Ratio Adj for Growth and Yield



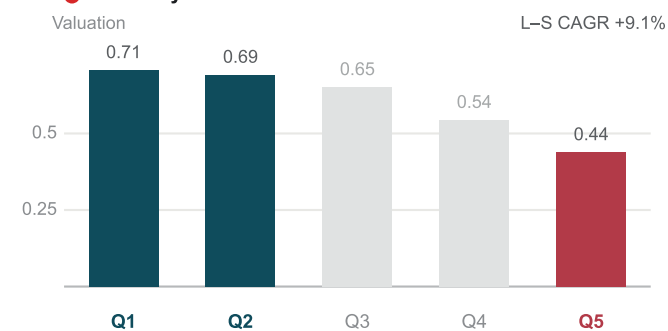
040 ● Trailing 12-Month Earnings/Trailing 12-Month Sales



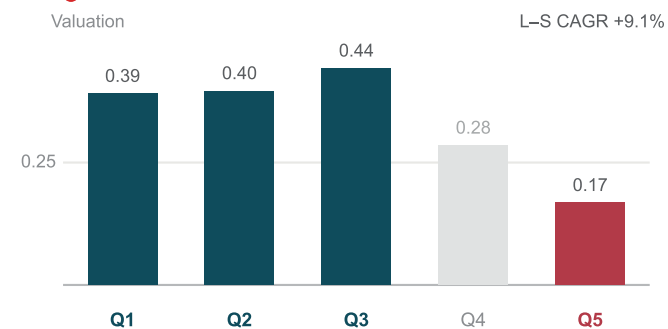
041 ● Net Cash Flow to Enterprise Value



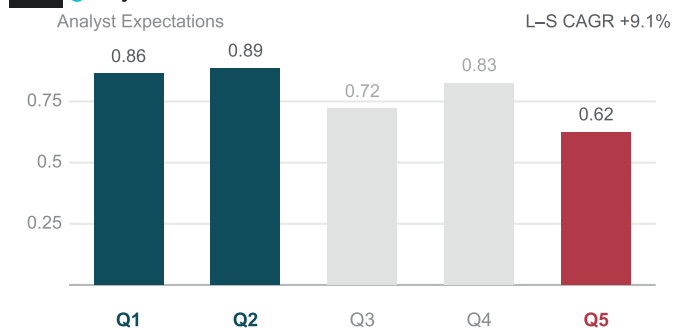
042 ● Solvency Ratio



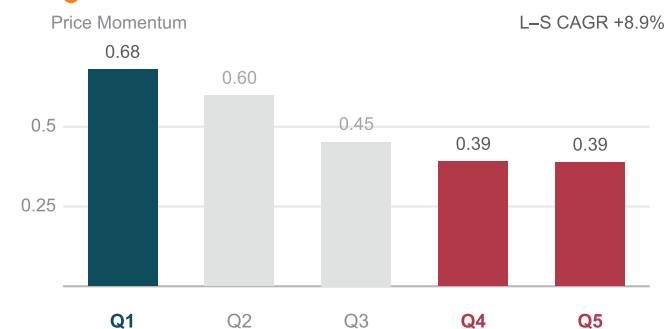
043 ● Dividends to Price Ratio



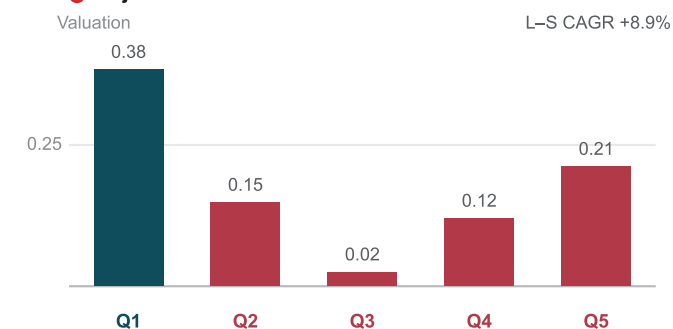
044 ● Adjusted Inverse PEG



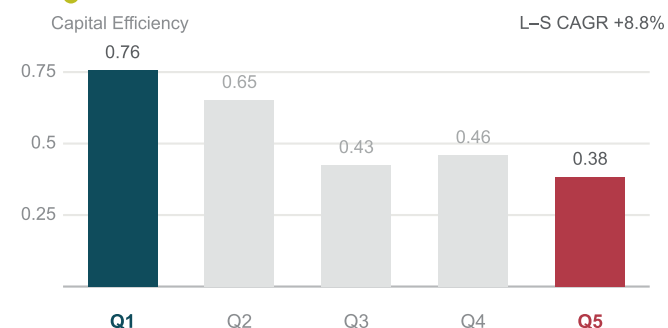
045 ● 180D Price TStat



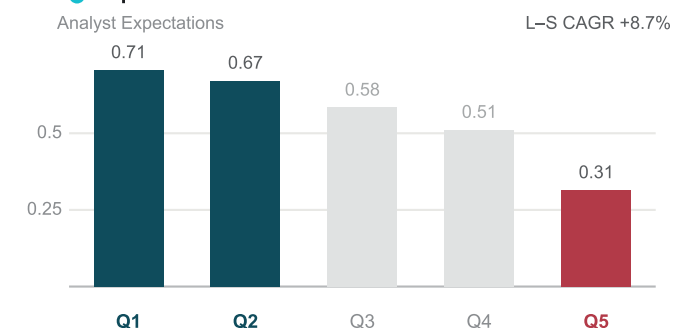
046 ● Adjusted EBIT to Price

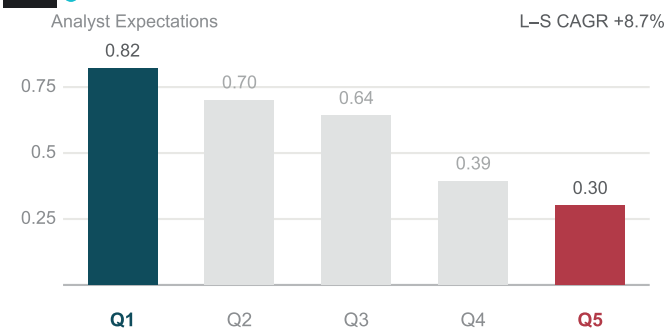
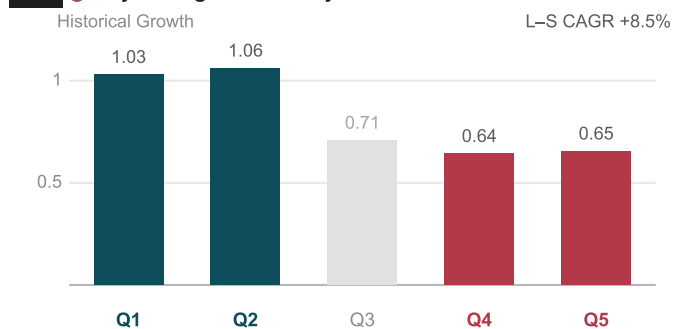
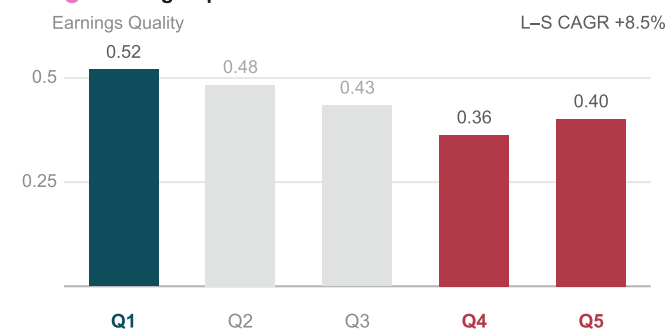
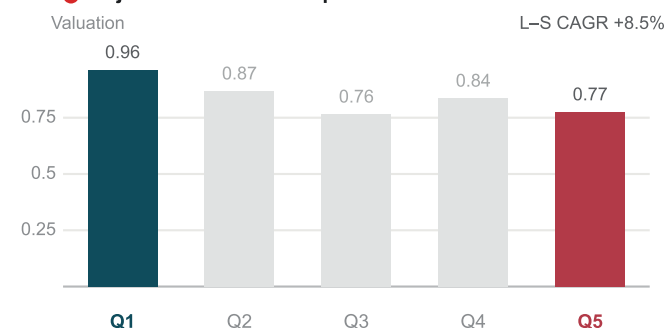
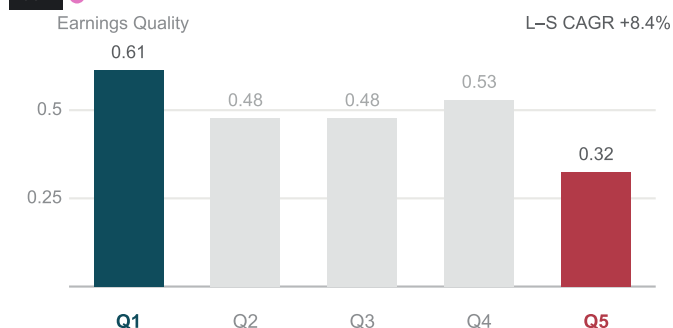


047 ● EBITDA to Assets



048 ● Expected LTG

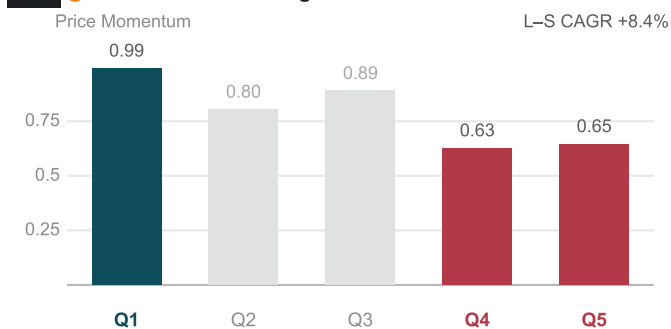


049 3M Revision in FY1 EPS Estimate**050 Adj 1Y Chg in Asset Adjusted FCF****051 Sales to EV Ratio****052 Working Capital to Sales****053 Adjusted Book to Enterprise Value****054 Balance Sheet Accruals**

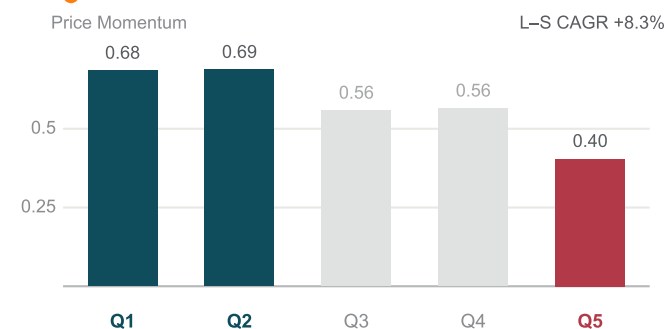
055 ● EBITDA to Enterprise Value



056 ● Short Interest Trading Volume



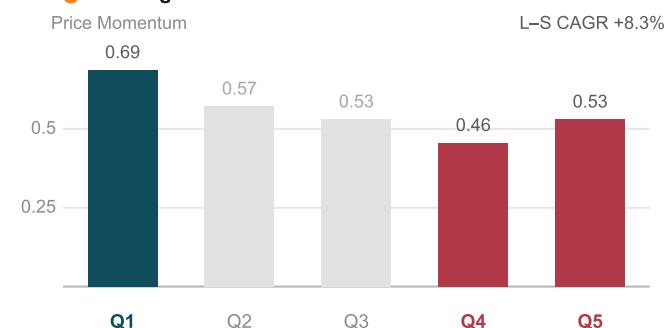
057 ● 12M - 1M Price Momentum



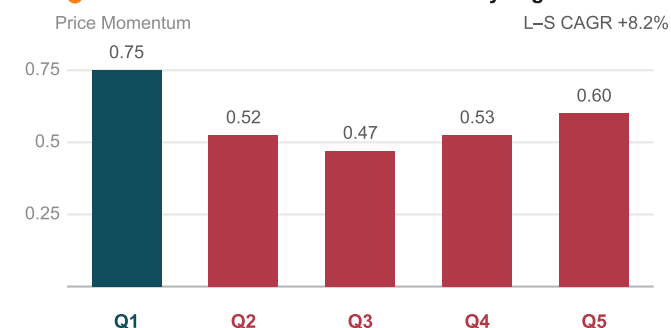
058 ● Amihud Illiquidity Measure



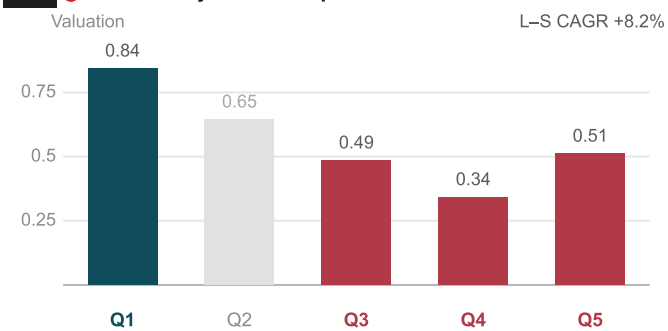
059 ● 52W High Low



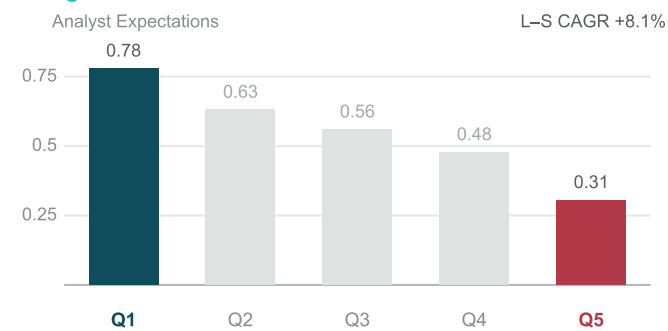
060 ● 52-Week Volume Price Trend with 20-Day Lag



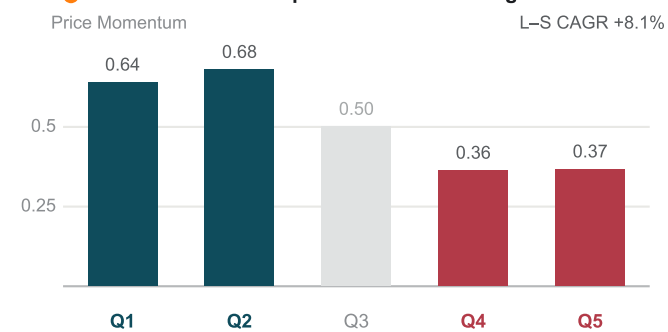
061 ● Cash to Adjusted Enterprise Value



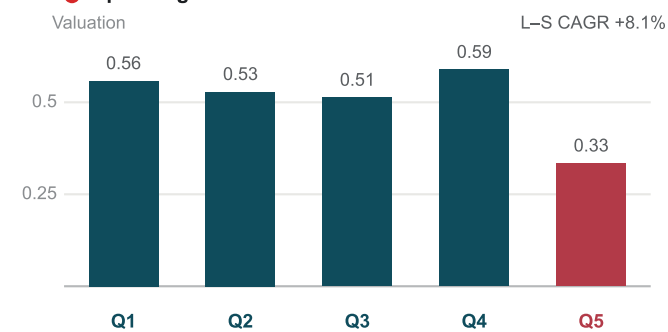
062 ● 3M Revision in FY2 EPS Estimate



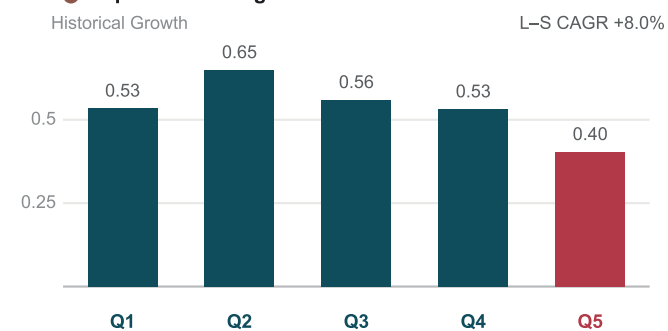
063 ● Institution Ownership Level - Active Manager



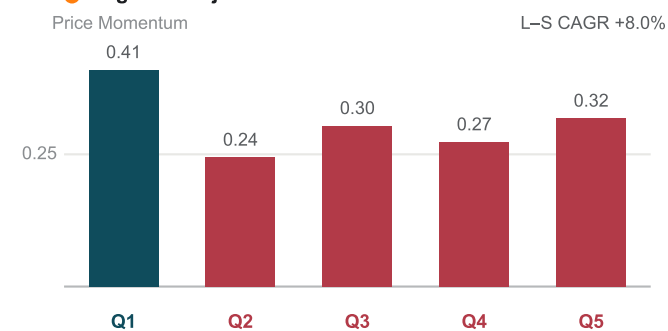
064 ● Operating Cash Flow to Price

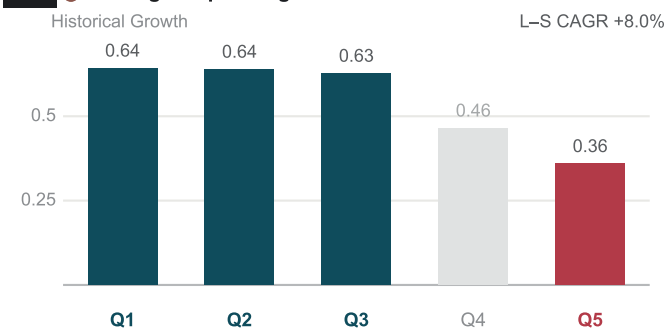
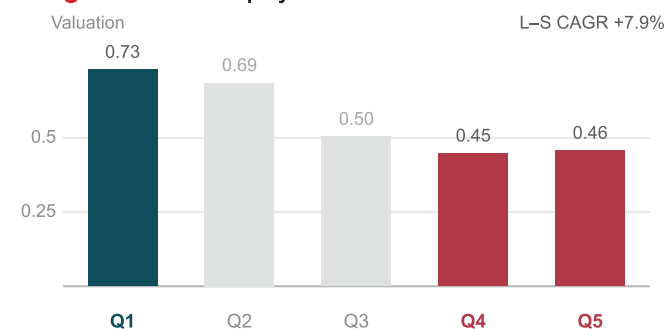
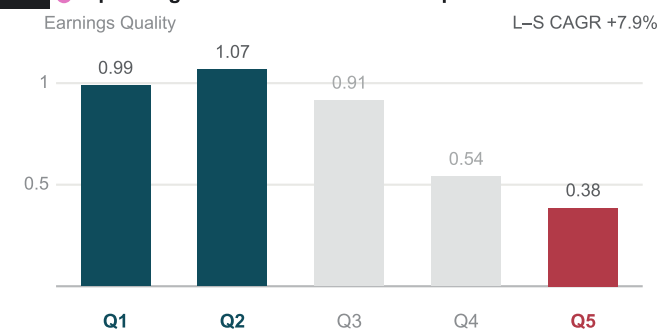
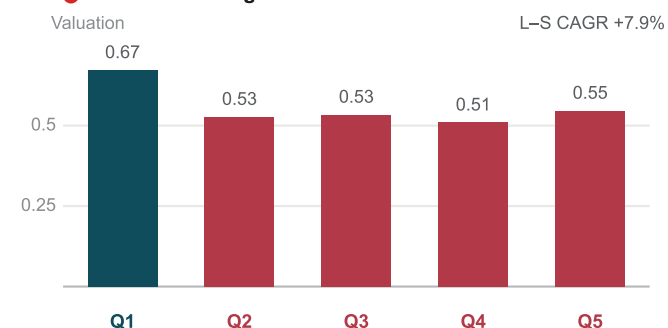


065 ● Expected Earnings Growth: Fiscal Year 2/Fiscal Year 1

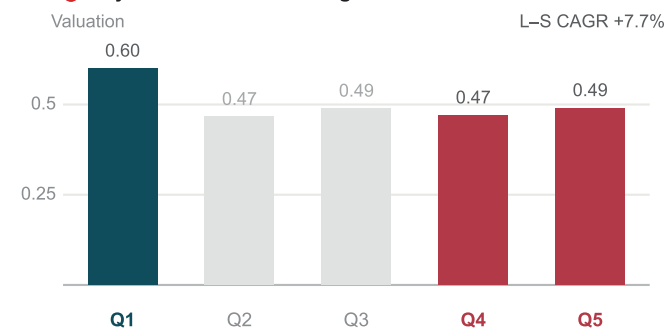


066 ● Log of Unadjusted Stock Price



067 ● 3Y Chg in Operating Cash Flow to Price**068 ● Street Revision Magnitude****069 ● Cash Flow to Equity****070 ● Operating Cash Flow to Invested Capital****071 ● Forward Earnings to Price****072 ● 12-Quarter Trendline in Trailing 12-Month Earnings**

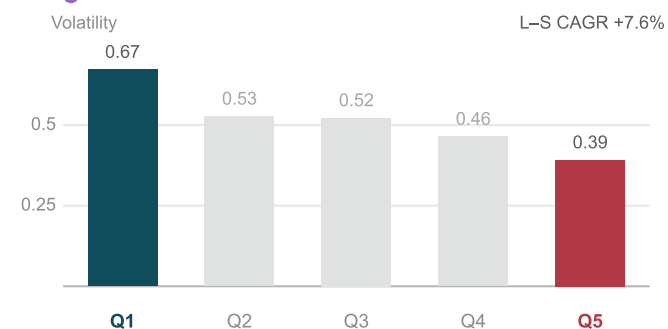
073 ● Adjusted Forward Earnings to Price



074 ● Gross Profit Margin



075 ● 60M CAPM Beta



076 ● Adjusted 3Y Chg in Asset Adjusted Free Cash Flow



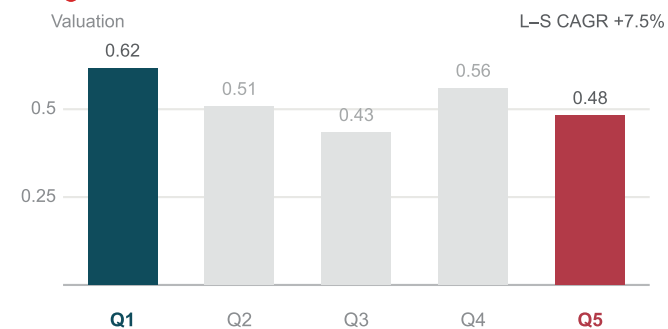
077 ● Adjusted Revision Magnitude



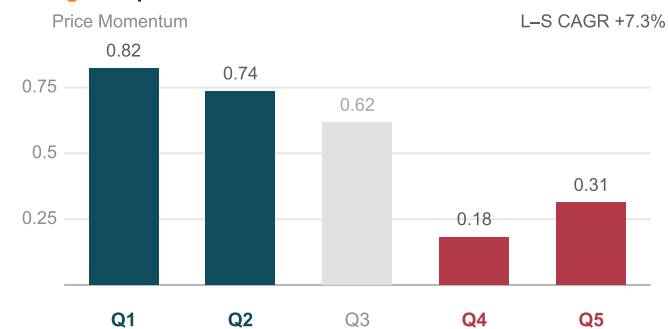
078 ● 1Y Chg in Asset Adjusted EPS



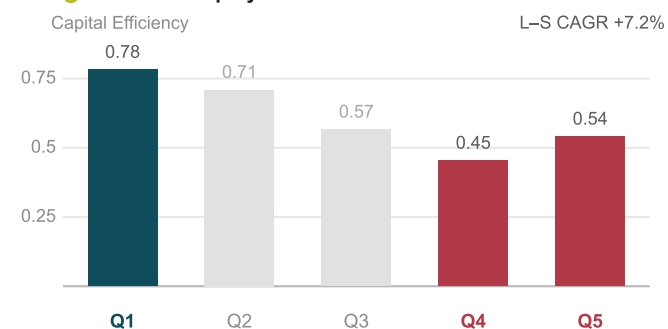
079 ● Net Cash Flow to Price



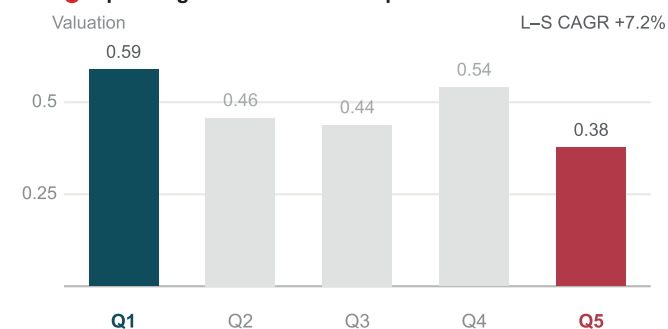
080 ● Sharpe Ratio



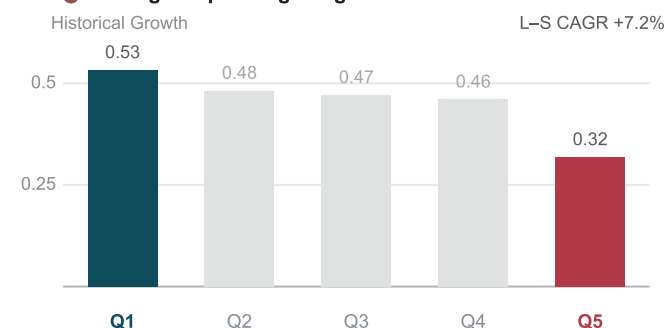
081 ● Return on Equity



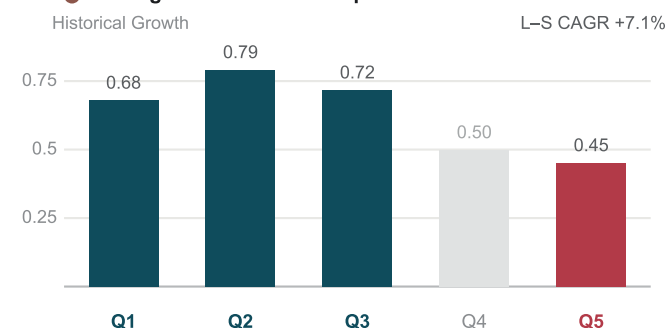
082 ● Operating Cash Flow to Enterprise Value

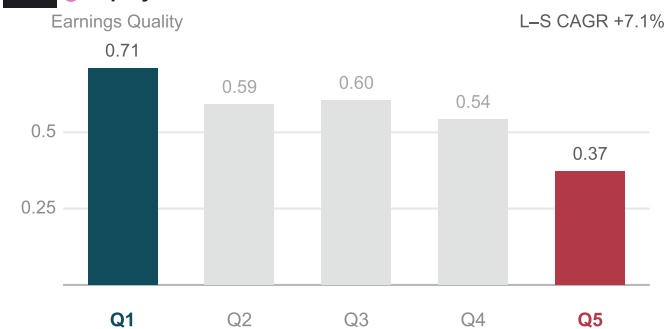
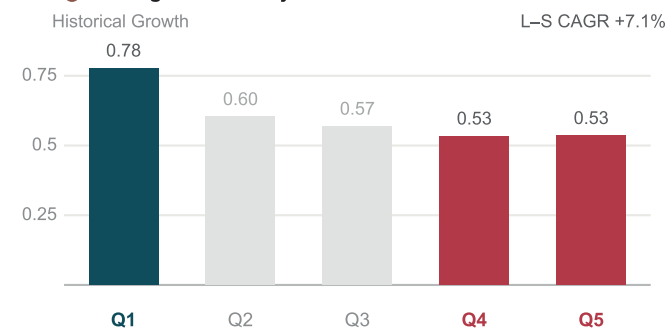
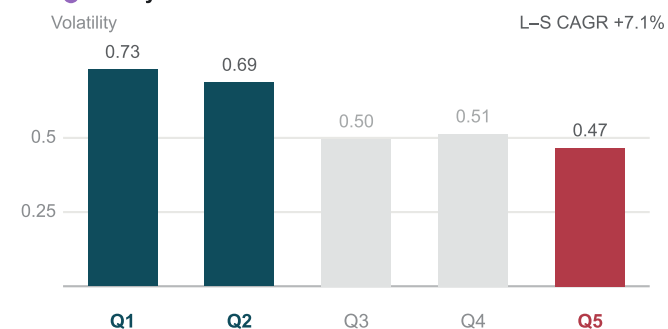


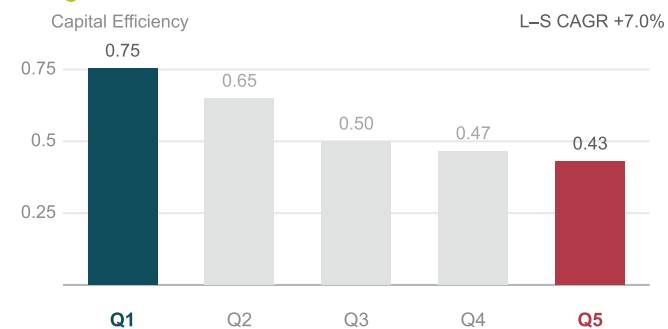
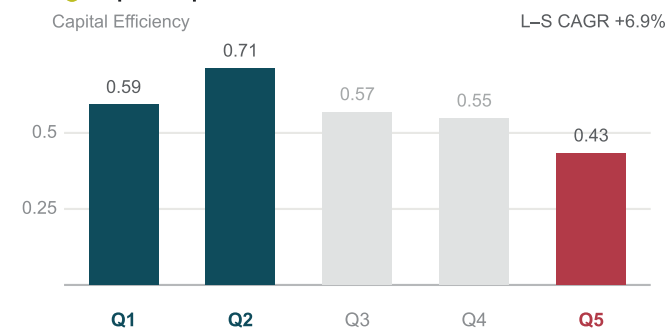
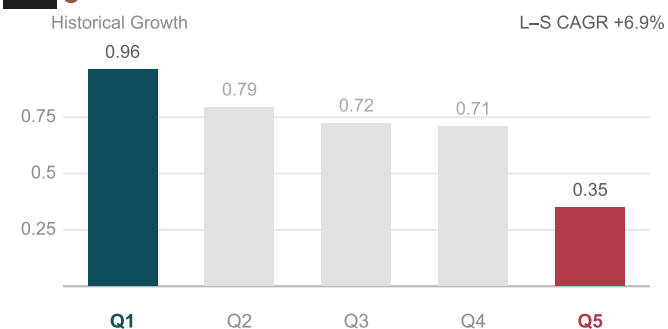
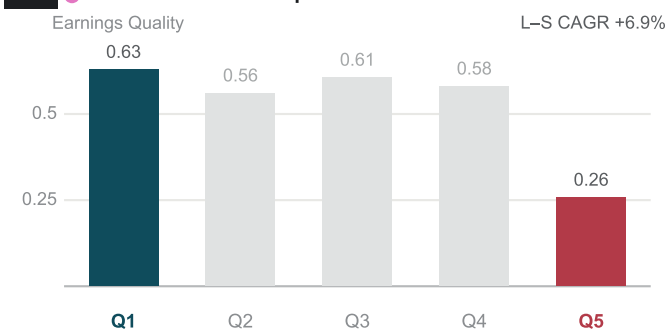
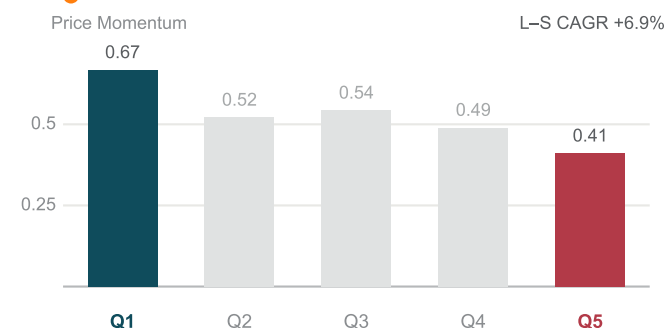
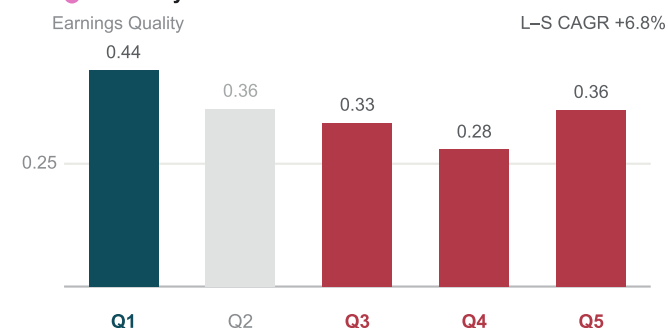
083 ● 1Y Chg in Operating Margin

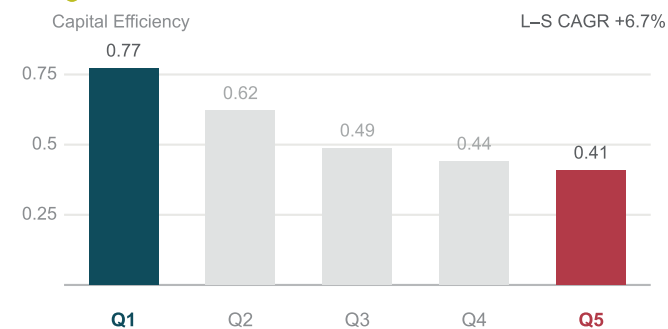
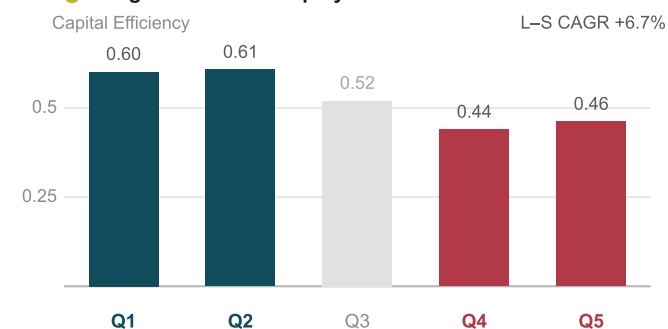
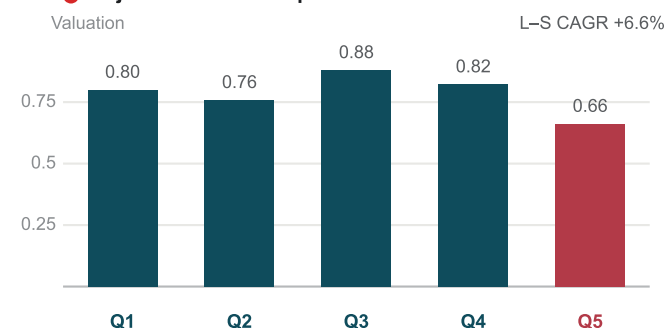
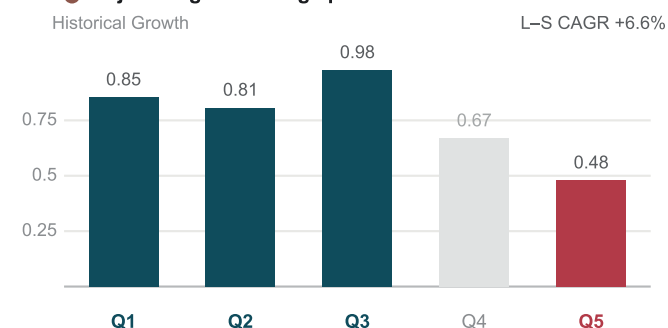


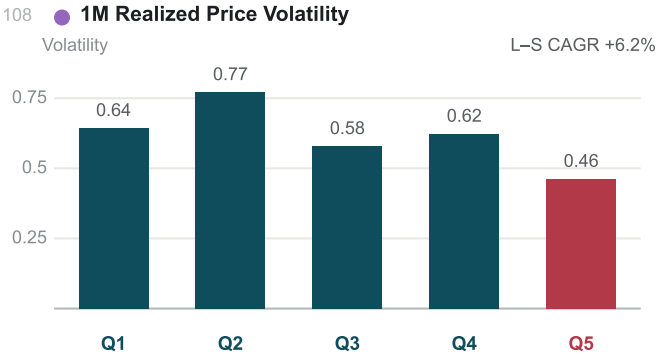
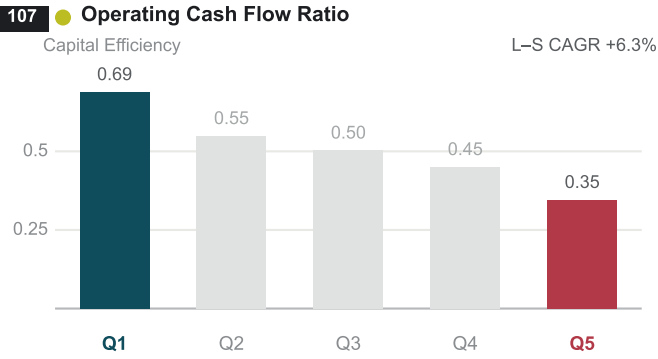
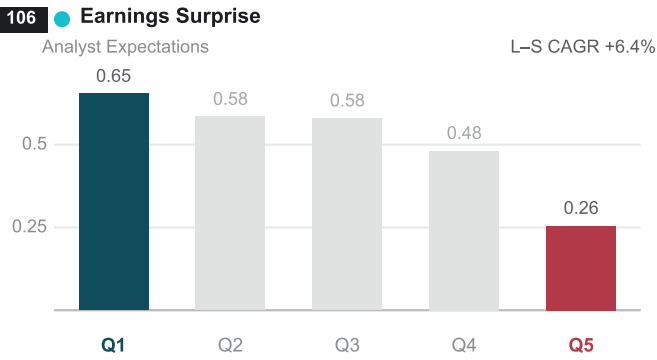
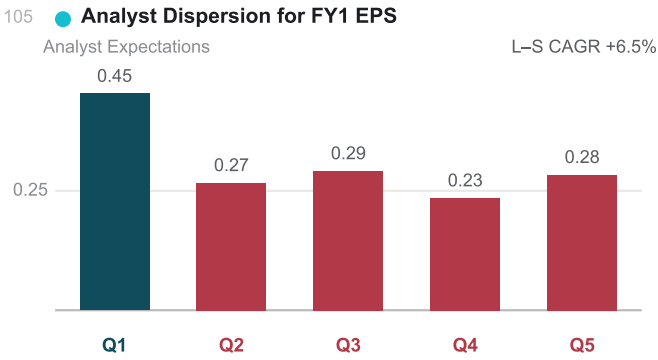
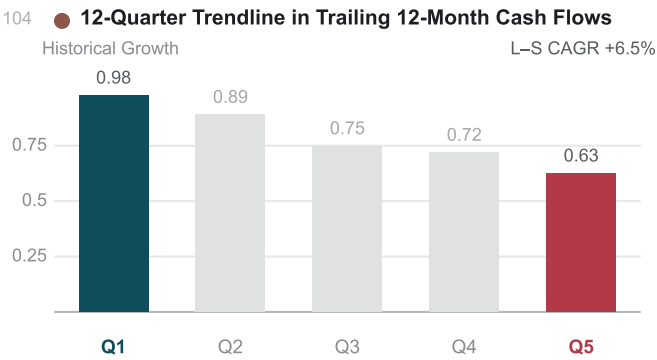
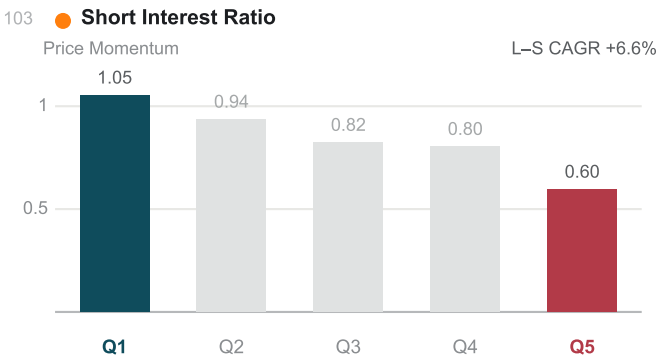
084 ● 1Y Chg in Free Cash Flow per Share



085 Equity Turnover**086 1Y Chg in Asset Adjusted Cash Flow****087 90 Day Coefficient of Variation****088 Working Capital to Assets Ratio****089 Inverse PEG****090 1Y Chg in Sales to Price**

091 ● **Normalized ROA**092 ● **Capital Expenditure to Total Assets**093 ● **Consecutive Quarters of Declines in TTM Overhead/Sales**094 ● **Sales to Invested Capital**095 ● **26W Price Stat**096 ● **Inventory Turnover**

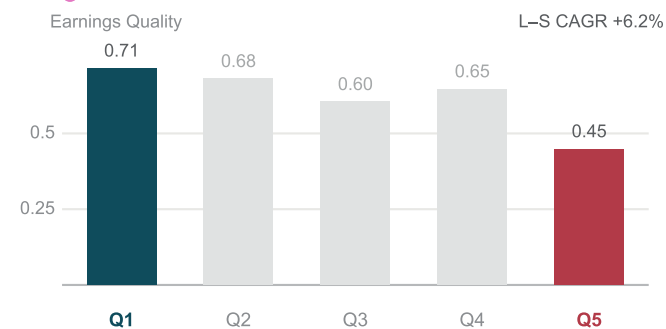
097 ● **Year over Year Change in SGA to Sales**098 ● **EBIT to Assets**099 ● **1Y Chg in Earnings to Price**100 ● **Long Term Debt to Equity Ratio**101 ● **Adj. EBITDA to Enterprise Value**102 ● **Adj 1Y Chg in Earnings per Share**



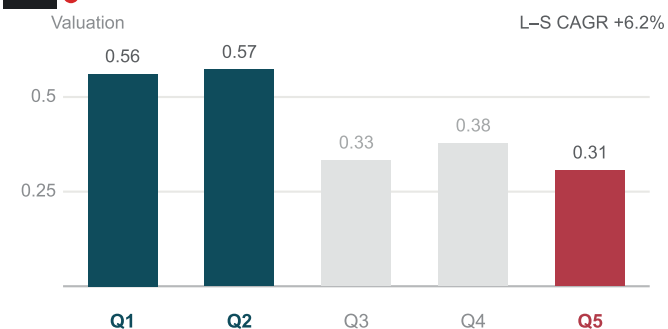
109 ● Volatility adjusted 12M return



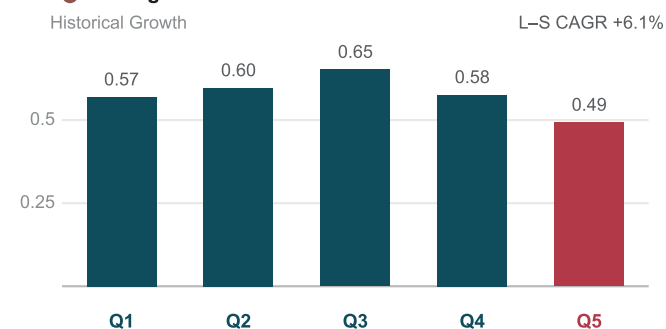
110 ● Current Ratio



111 ● Cash to EV Ratio



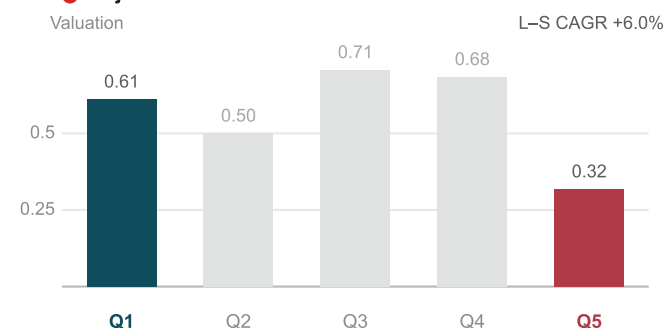
112 ● 1Y Chg in Cash Flow to Price

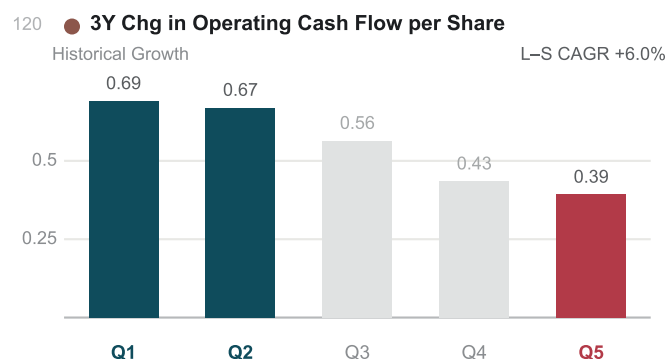
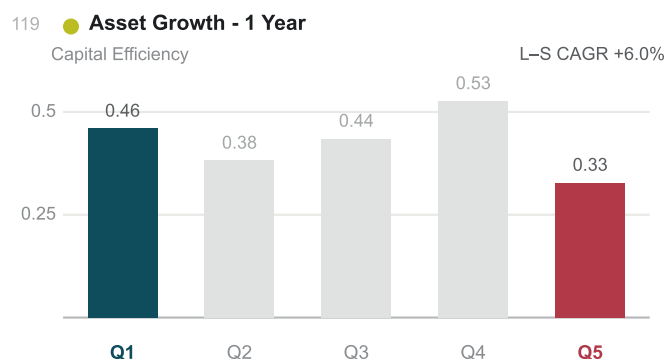
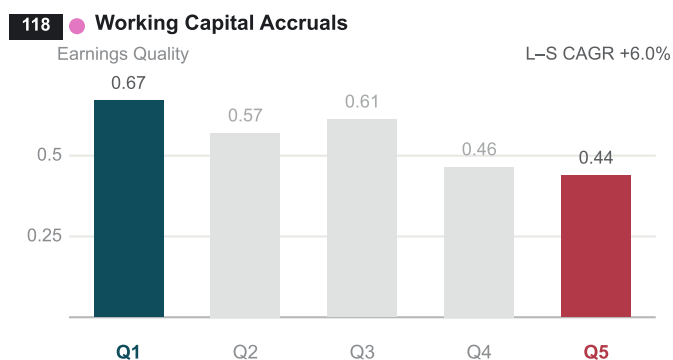
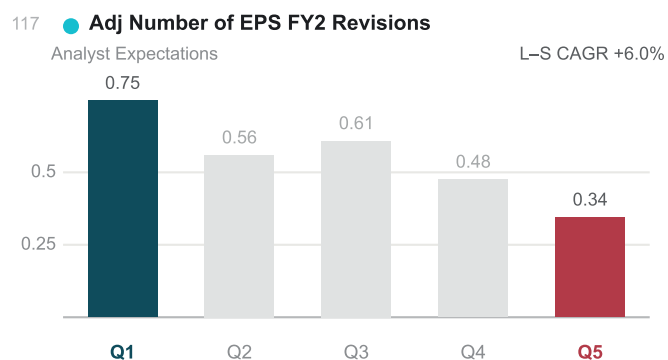
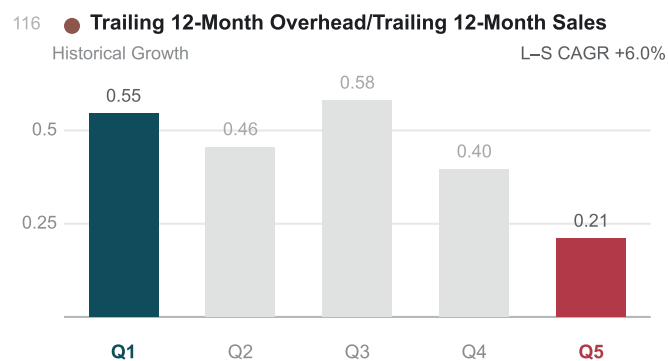
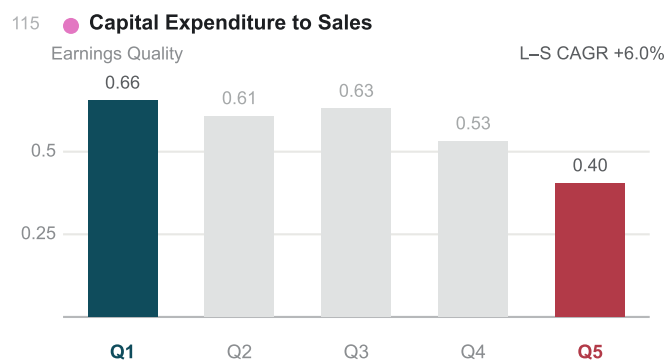


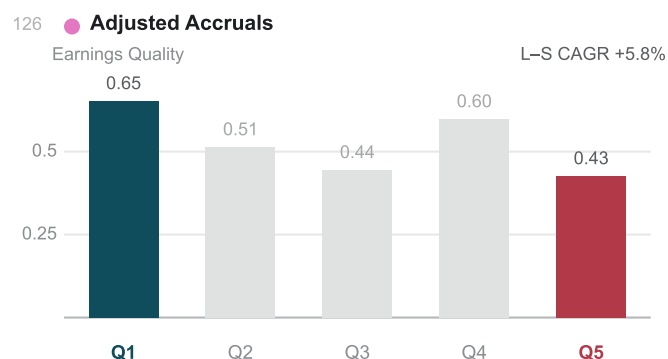
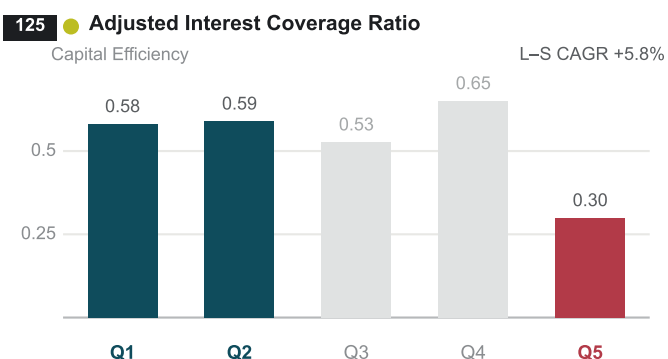
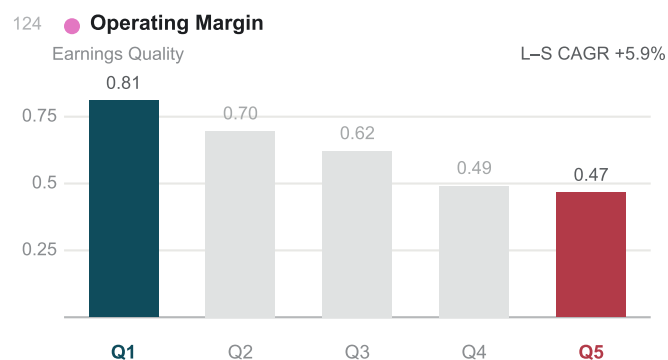
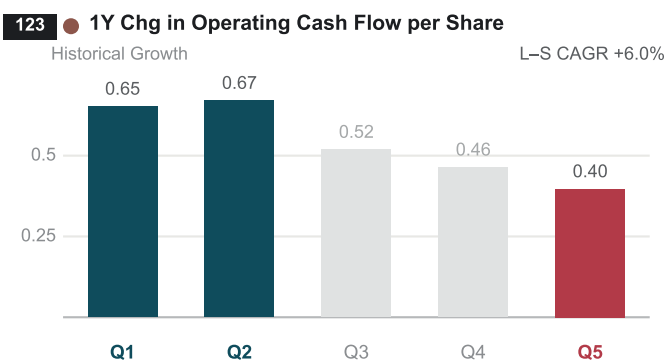
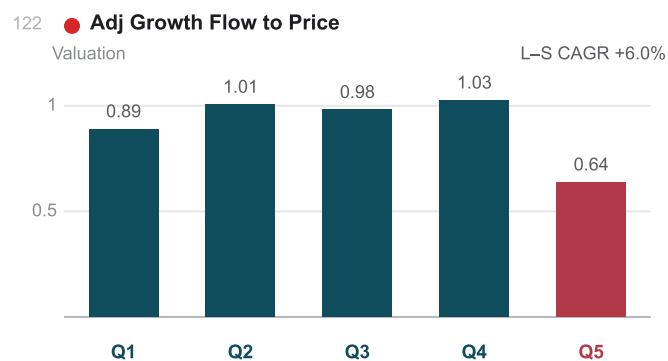
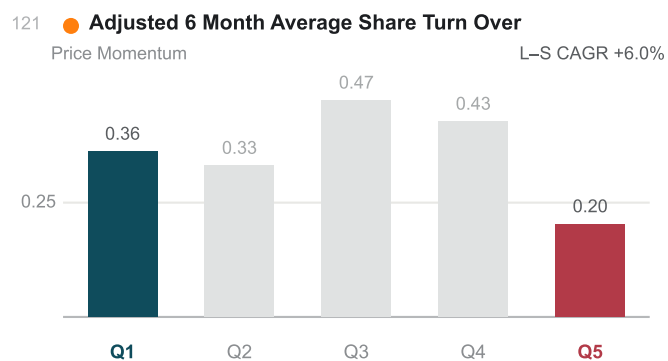
113 ● EBITDA Margin

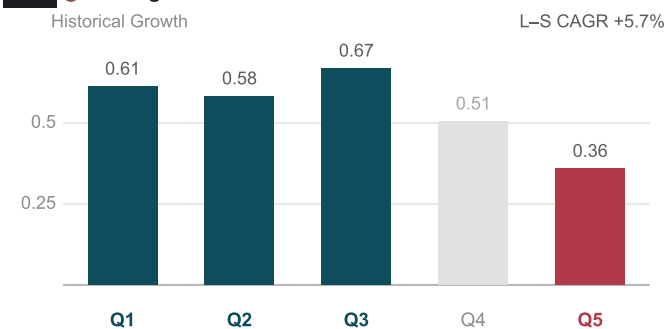
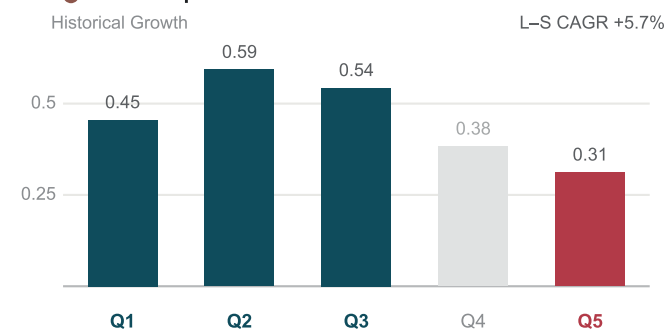
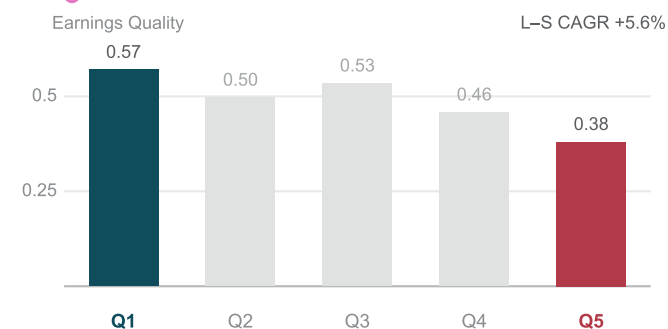
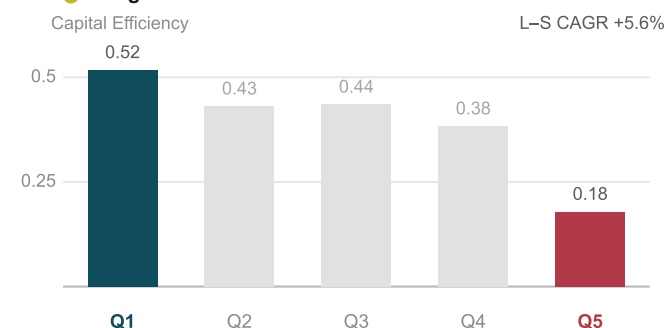
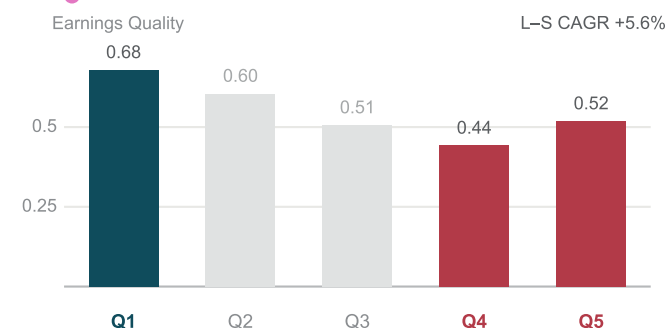


114 ● Adj Free Cash Flow to Price







127 ● 3Y Chg in Free Cash Flow to Price**128 ● Ownership Breadth Stability of Institution Ownership by Buyers****129 ● Dividend per Share Growth****130 ● Accrual Ratio - Balance Sheet****131 ● Long Term Debt to Assets Ratio****132 ● Accrual Ratio - Cash Flows**

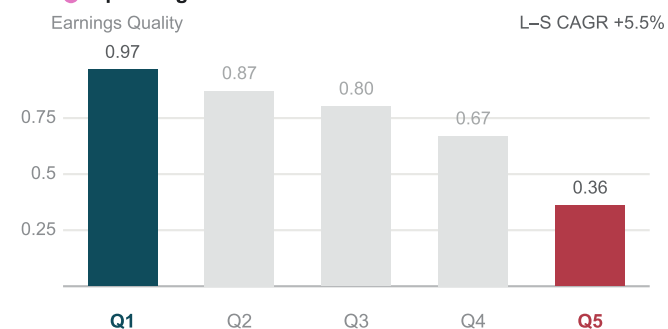
133 ● Year over Year Change of EPS to Sales



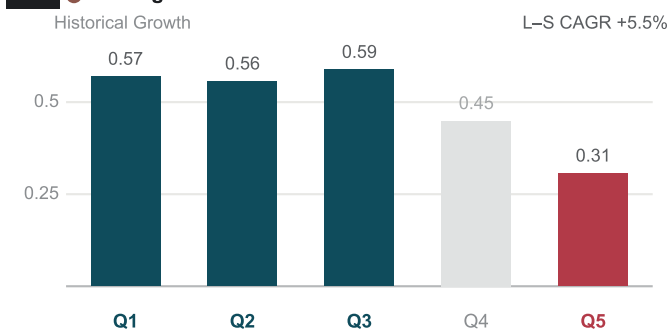
134 ● Free Cash Flow to Enterprise Value



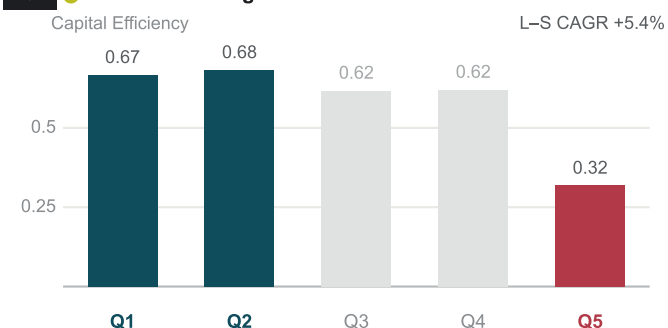
135 ● Operating Cash Flow to Asset



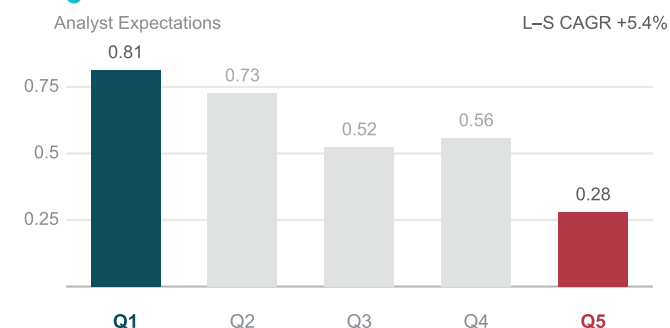
136 ● 3Y Chg in Sales to Price

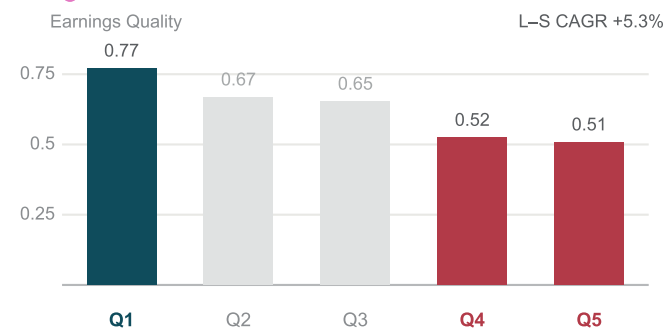
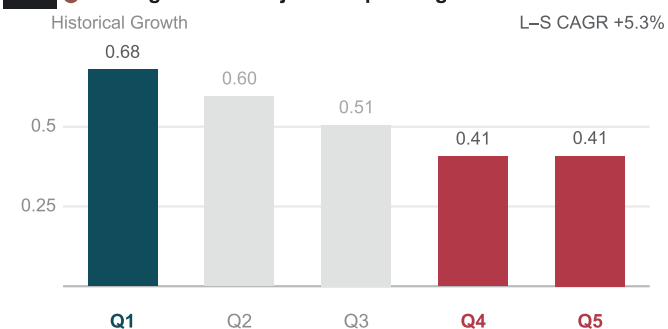
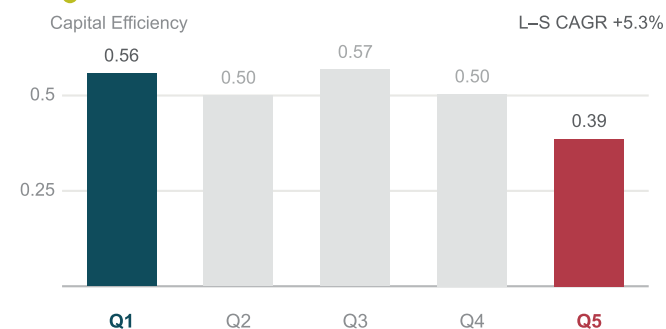
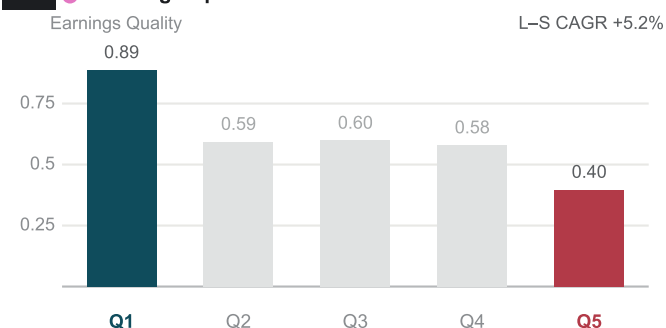


137 ● Interest Coverage Ratio



138 ● Number of EPS FY1 Revisions



139 ● Num of Quarters of Declines in Receivables and Inventories/...**140 ● Quick Ratio****141 ● 1Y Chg in Asset Adjusted Operating Cash Flow****142 ● Debt to Assets Ratio****143 ● Forward Free Cash Flow to Price Ratio****144 ● Working Capital to Total Assets**

145 ● Net Income Stability



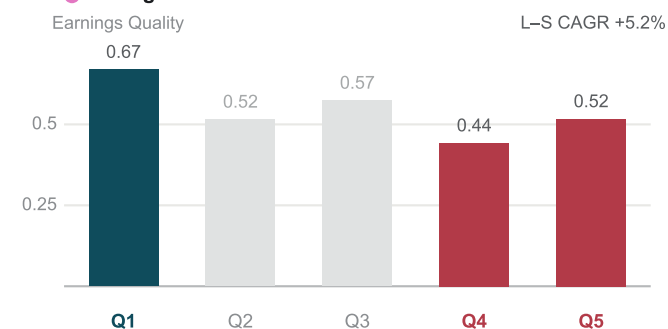
146 ● Return on Equity Stability



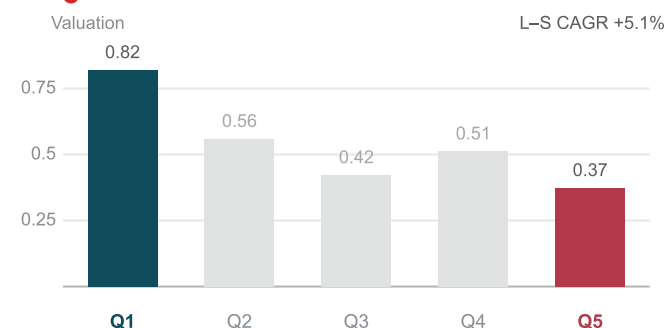
147 ● 1Y Chg in Asset Adjusted Free Cash Flow



148 ● Change EPS to Sales

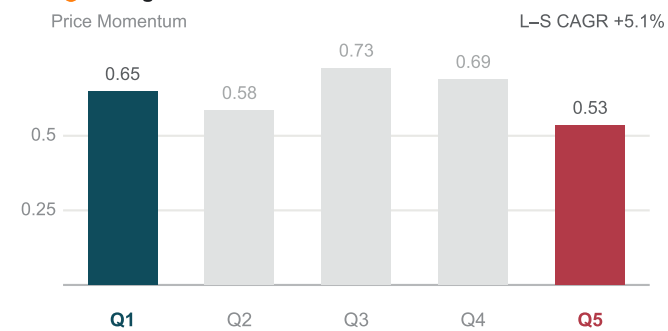
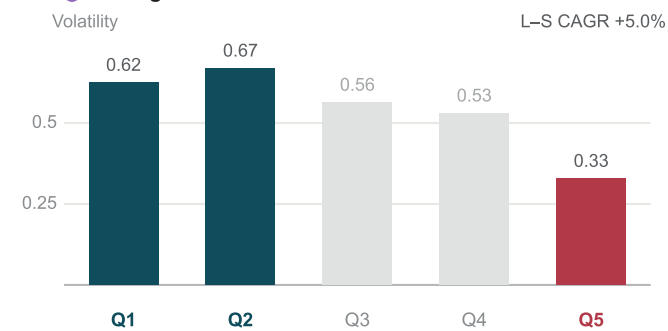
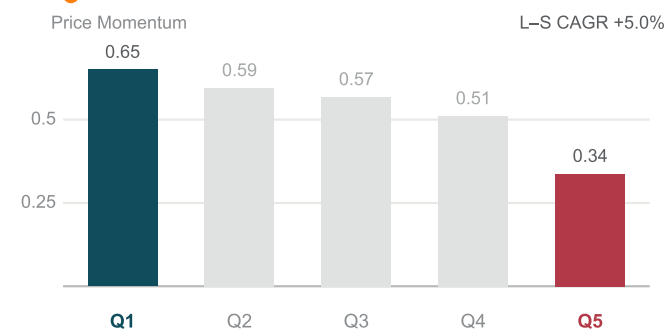
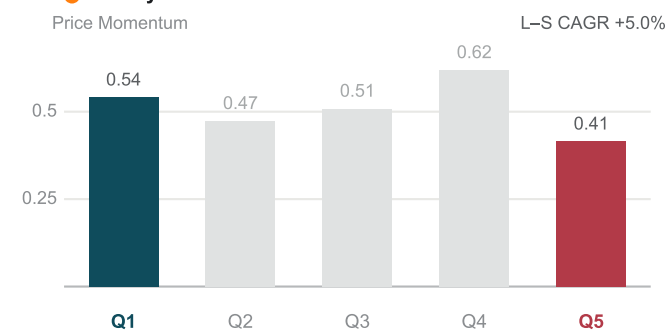


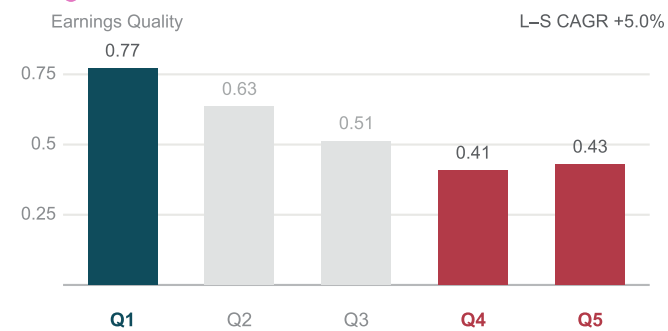
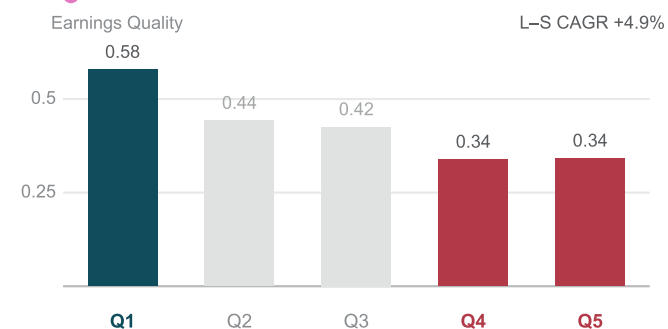
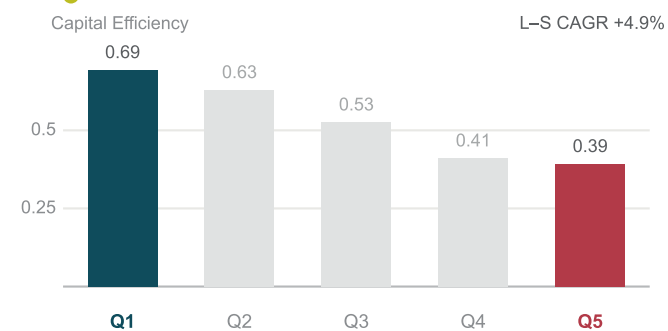
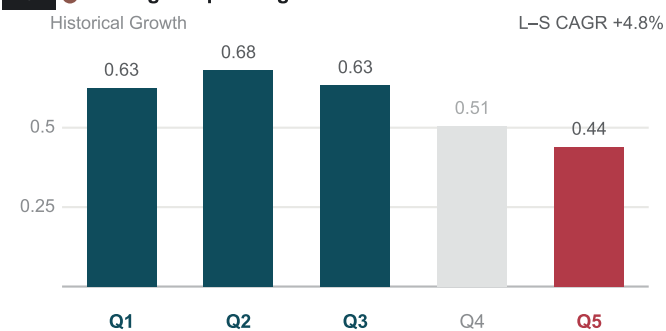
149 ● Cash to Total Assets



150 ● 6M Momentum in Trailing 12M Sales



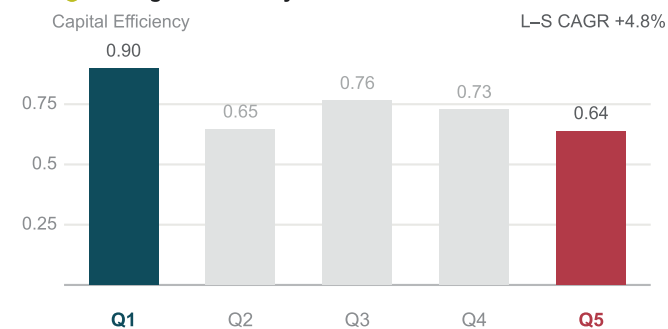
151 ● **Change in Number of Holders**152 ● **1Y Chg in Amihud**153 ● **24M Residual Return Variance**154 ● **Closing Price to 52 Week High**155 ● **Free Cash Flow Stability**156 ● **30 Day Coefficient of Variation of Volume to Price**

157 ● **Cash Ratio**158 ● **3Y Chg in Asset Adjusted Operating Cash Flow**159 ● **SG&A to Sales**160 ● **Return on Assets Stability**161 ● **Receivables Turnover Ratio**162 ● **1Y Chg in Operating Cash Flow to Price**

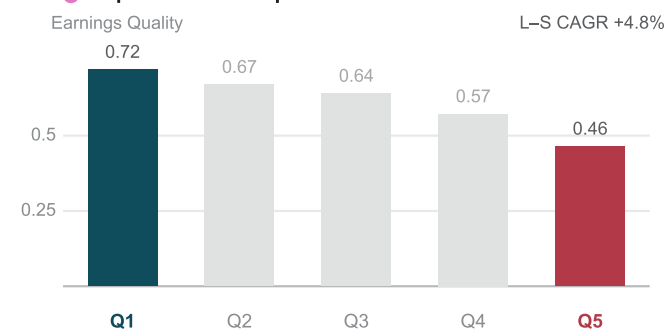
163 ● 1Y Chg in Shares Outstanding



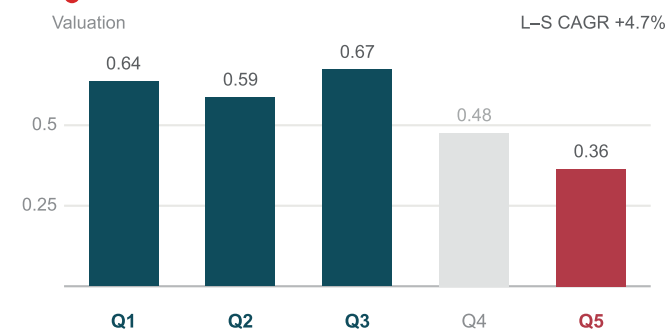
164 ● 1Y Chg in Stock Buybacks



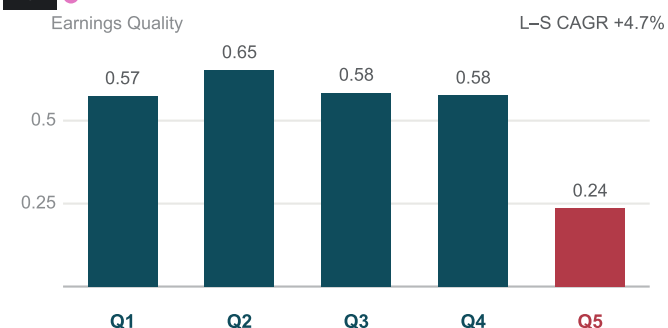
165 ● Depreciation to Capex Ratio



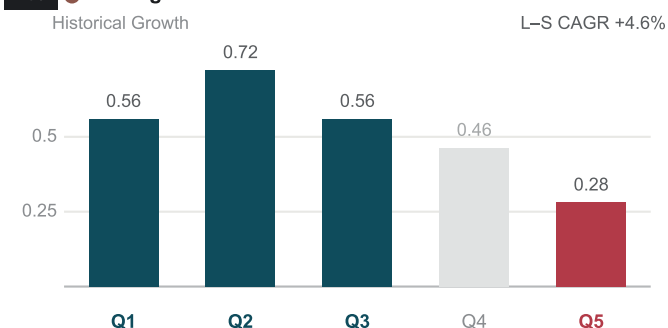
166 ● Free Cash Flow to Price

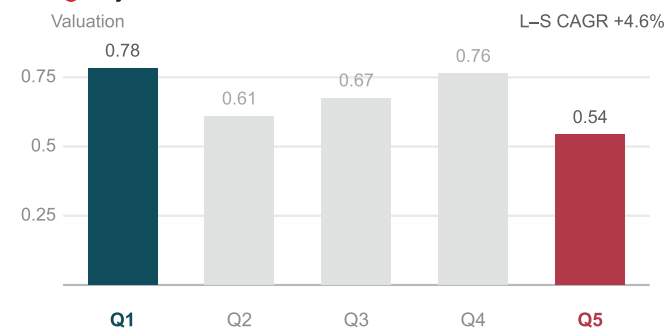
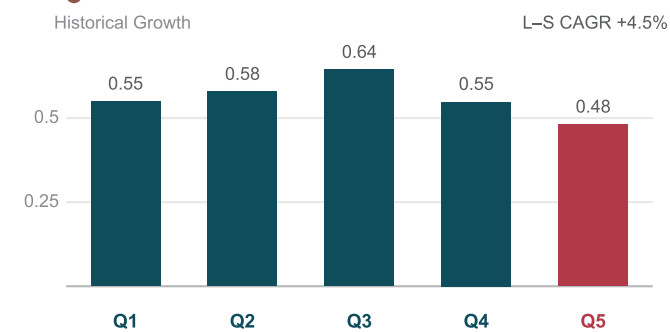
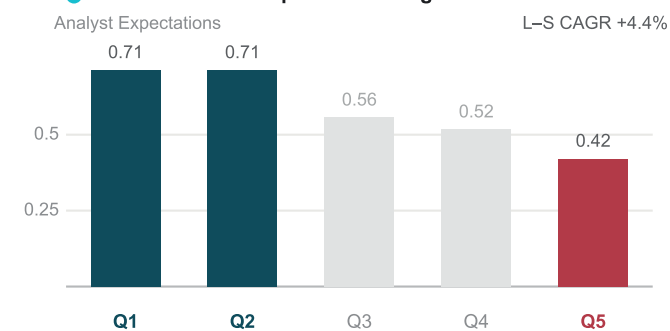


167 ● Asset Turnover

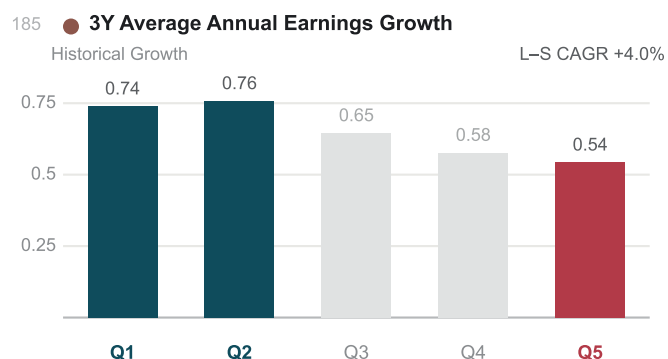
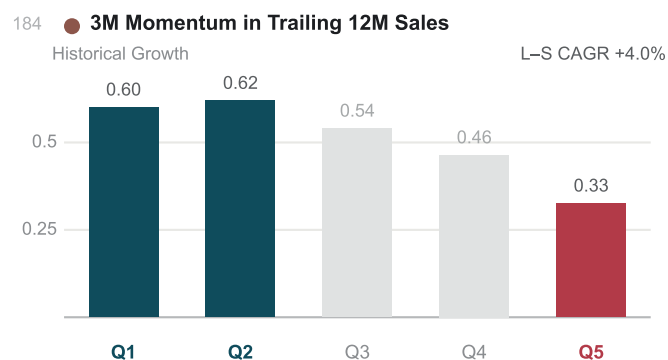
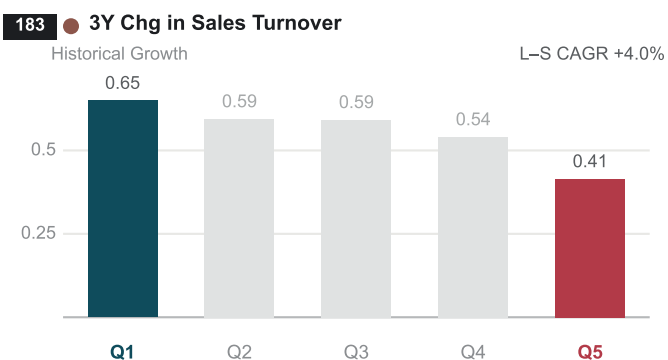
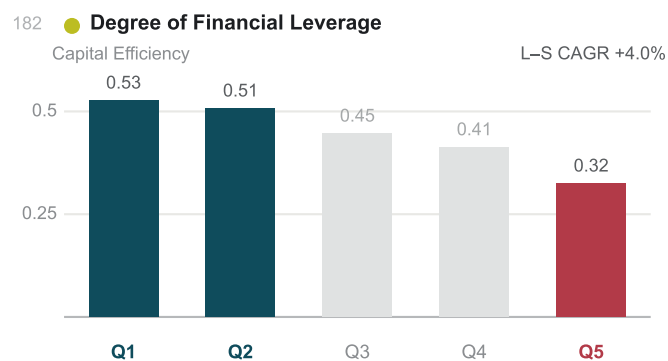
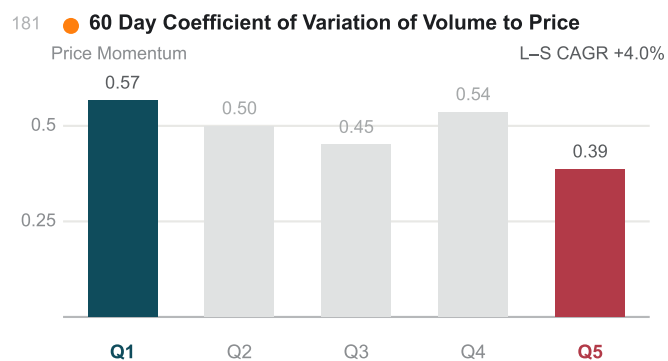


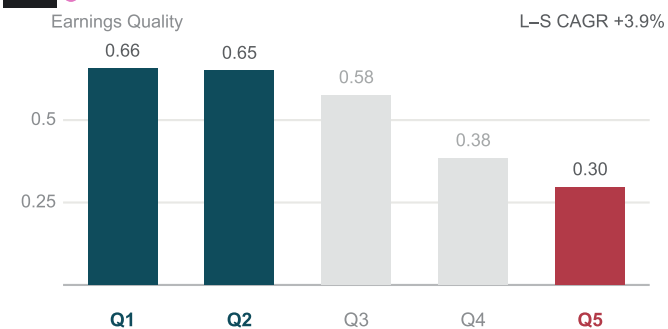
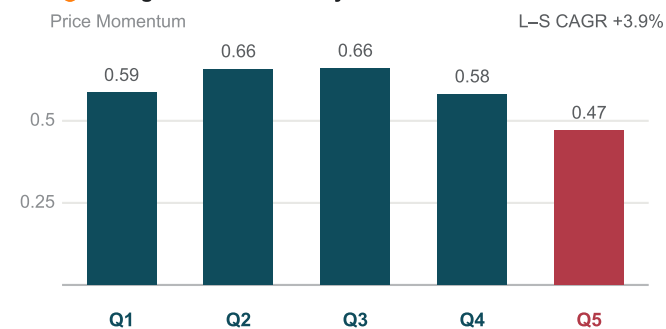
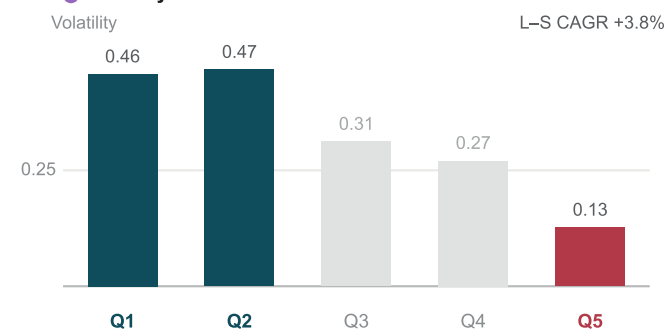
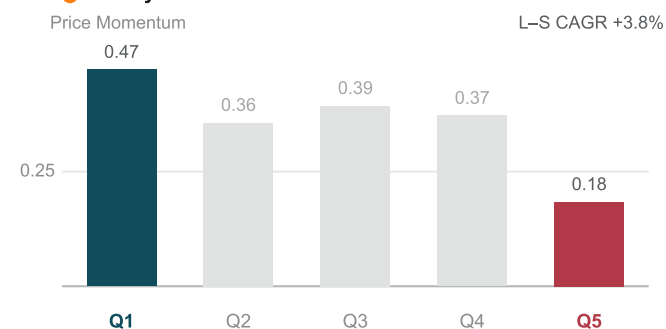
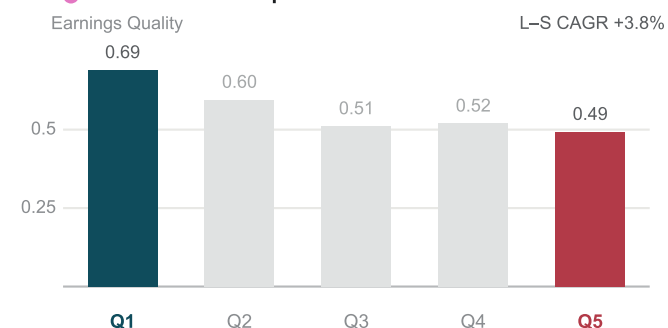
168 ● 1Y Chg in Free Cash Flow to Price



169 ● **Adj Forward Free Cash Flow to Price Ratio**170 ● **EPS Stability**171 ● **Cash Flow to Invested Capital**172 ● **Dividend Growth**173 ● **Return on Invested Capital**174 ● **Standardized Unexpected Earnings**



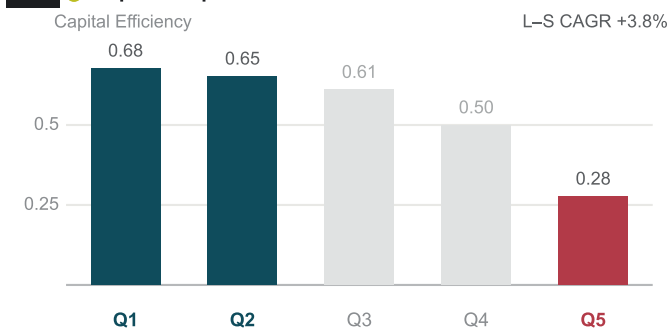


187 ● Free Cash Flow to Sales

188 ● Change in Number of Buyers

189 ● 130 Day Minimum Return

190 ● 10 Day MACD Trend

191 ● ROA 60 Month Slope

192 ● Cash Conversion Cycle

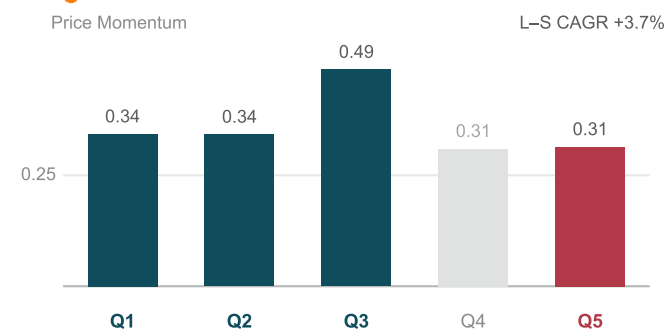

193 ● Net Profit Margin



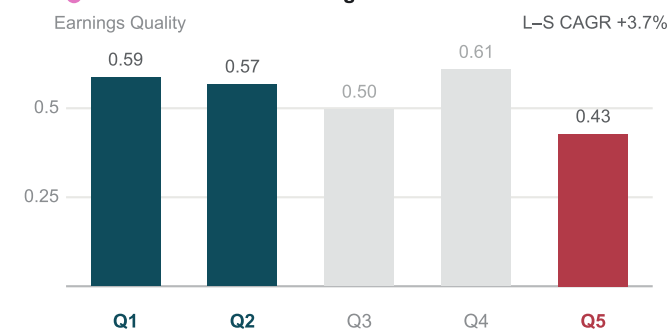
194 ● Capital Acquisition Ratio



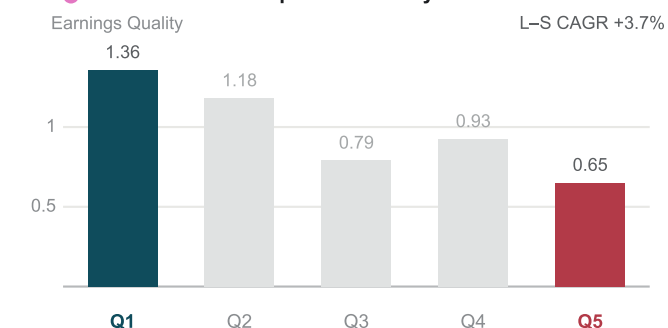
195 ● Share Turnover



196 ● Sales to Gross Profit Margin Growth

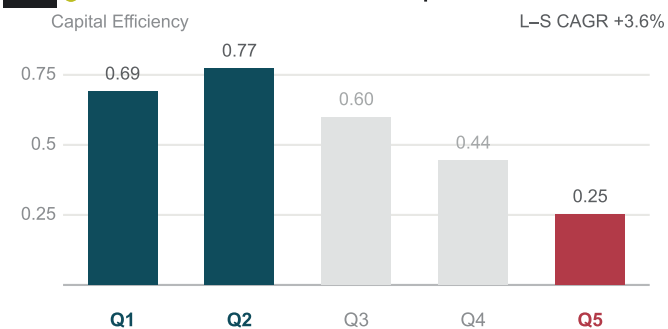
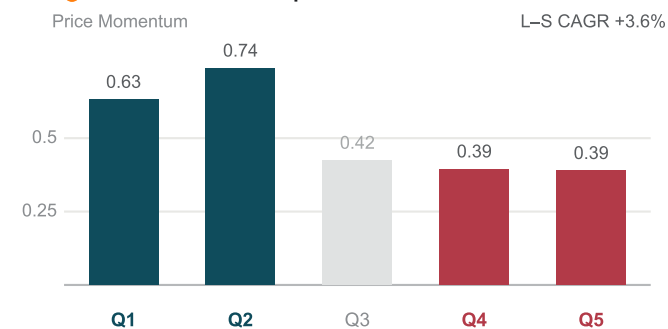
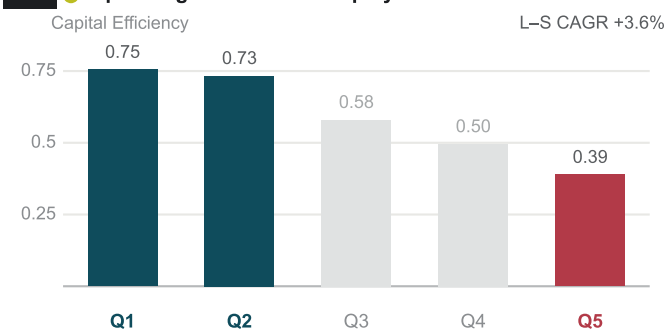
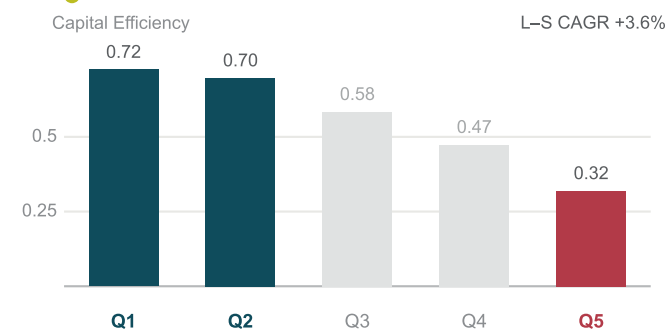
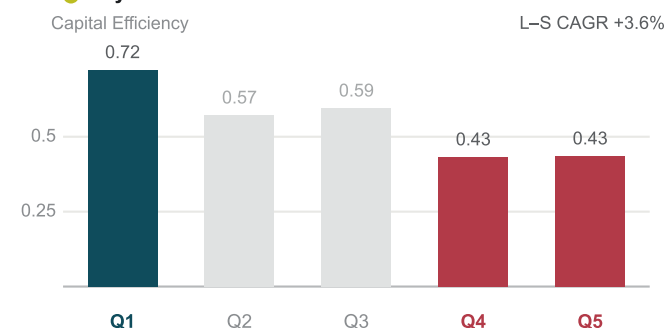
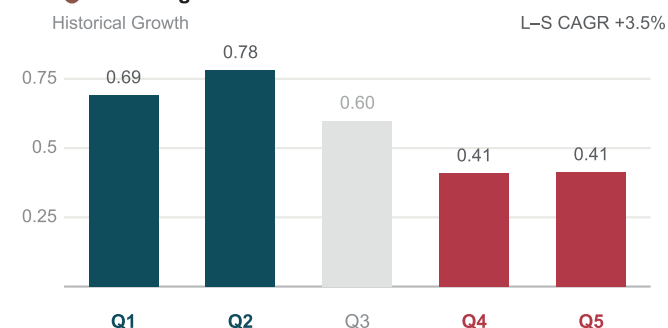


197 ● Research & Development Intensity



198 ● Asset Adjusted Capital Investments

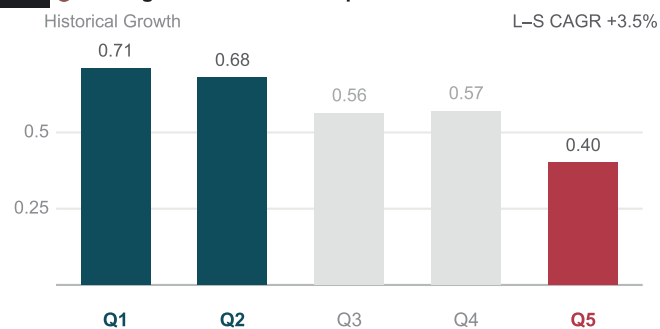


199 ● Cash Flow Return on Invested Capital**200 ● Institution Ownership Level****201 ● Operating Cash Flow to Equity****202 ● Gross Profit to Assets****203 ● Payout Ratio****204 ● 3Y Average Annual Sales Growth**

205 ● 3Y Chg in Asset Adjusted EPS



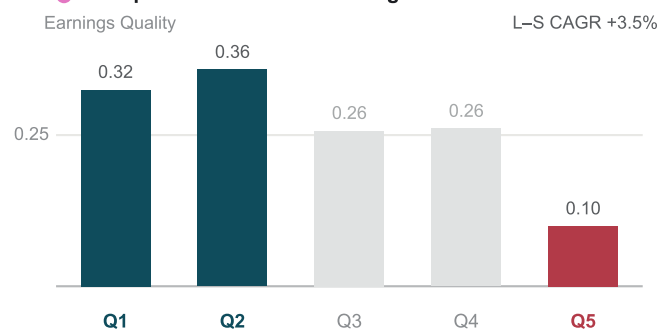
206 ● 3Y Chg in Free Cash Flow per Share



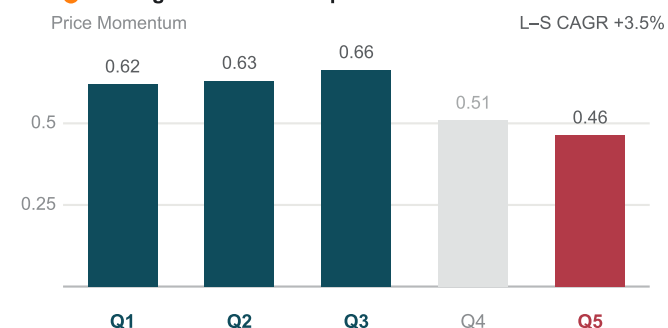
207 ● 1Y Chg in Earnings per Share



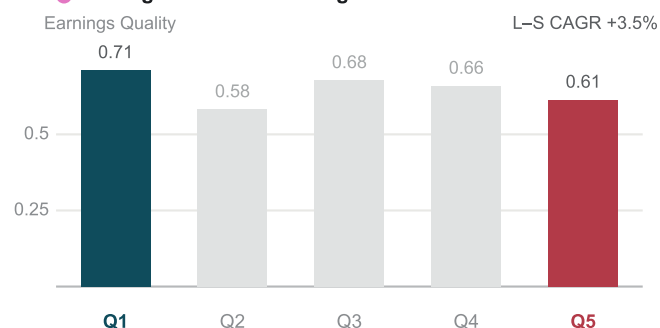
208 ● Unexpected Receivables Change

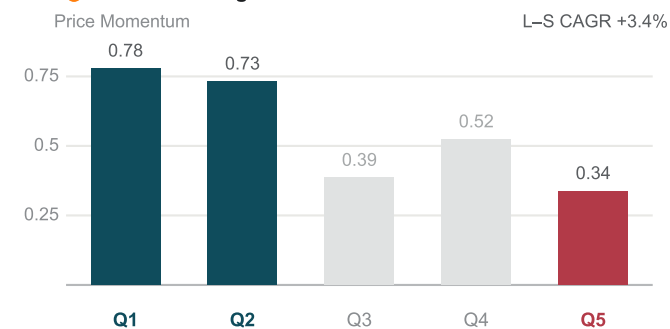
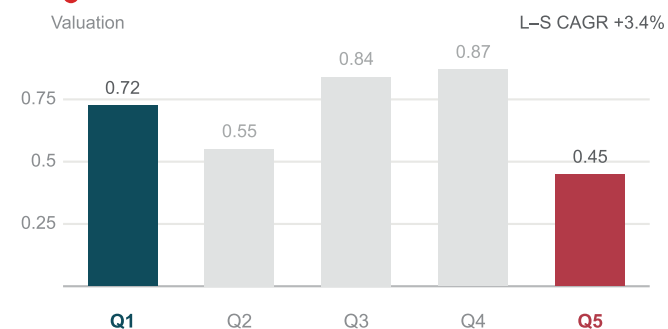
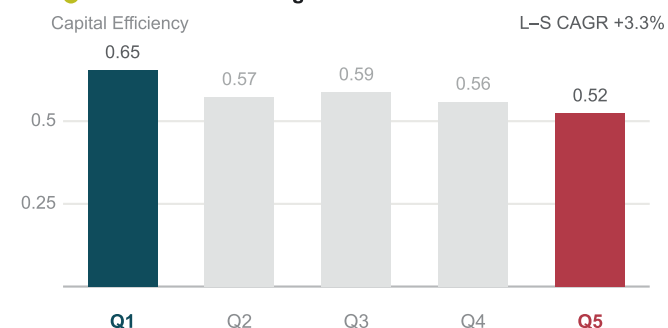
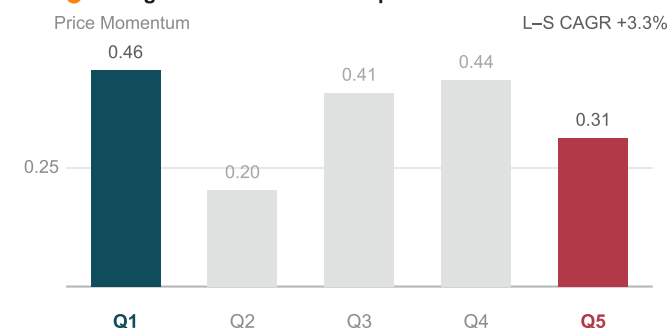


209 ● 6M Chg in 36M CAPM Alpha



210 ● 1Y Chg in Sales to Earnings

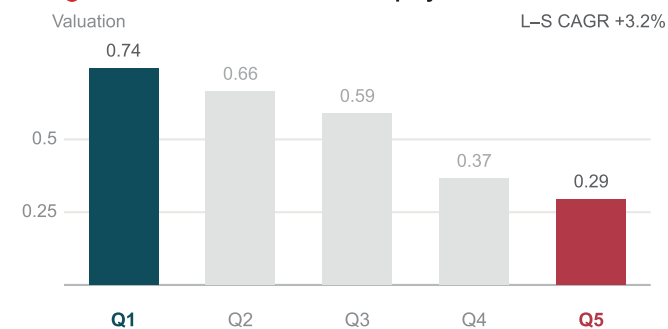


211 ● Operating Cash Flow Stability**212 ● 6 Month Average Share Turn Over****213 ● EBIT to Price****214 ● Ownership Breadth Stability of Institution Ownership by Hol...****215 ● Year over Year Change of Total Debt****216 ● Foreign Institution Ownership**

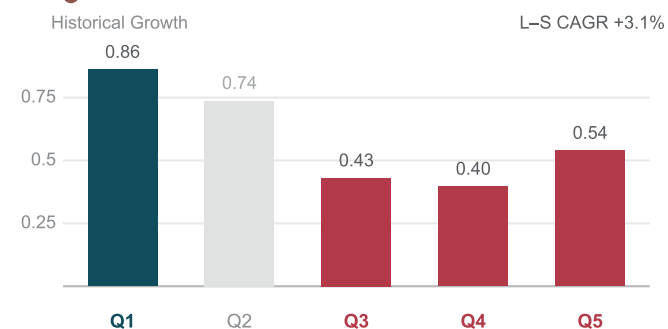
217 ● 3Y Chg in Asset Adjusted Free Cash Flow



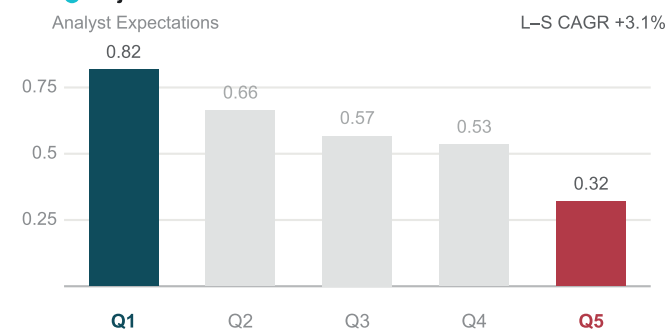
218 ● Free Cash Flow to Common Equity



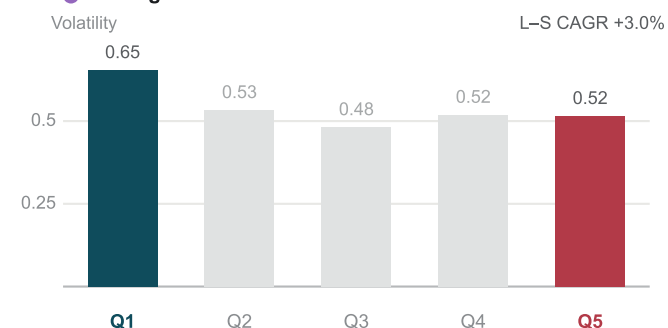
219 ● Sustainable Growth Rate



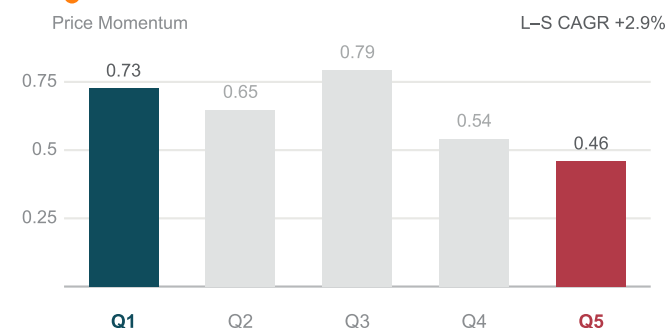
220 ● Adjusted Number of EPS FY1 Revisions



221 ● 1Y Chg in Debt to Assets



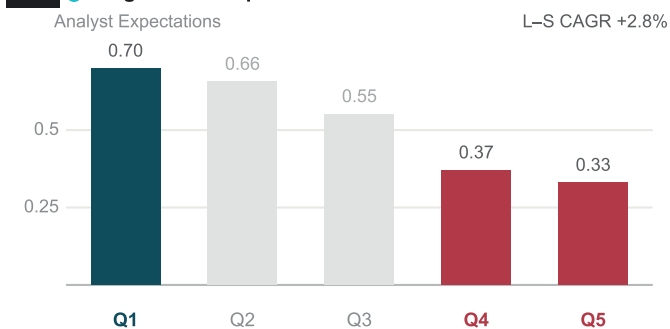
222 ● Standardized 36M Residual



223 ● 1Y Chg in Cash Flow per Share



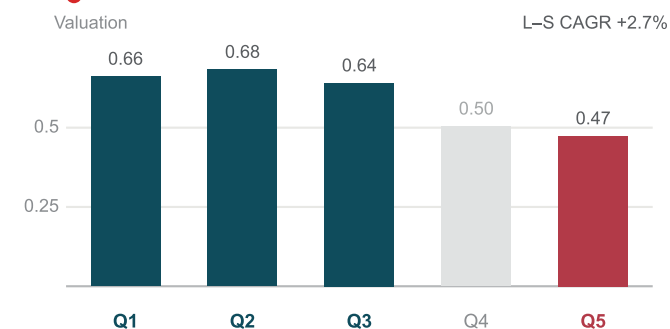
224 ● Target Price Gap to 6M EMA



225 ● 3Y Chg in Asset Adjusted Cash Flow



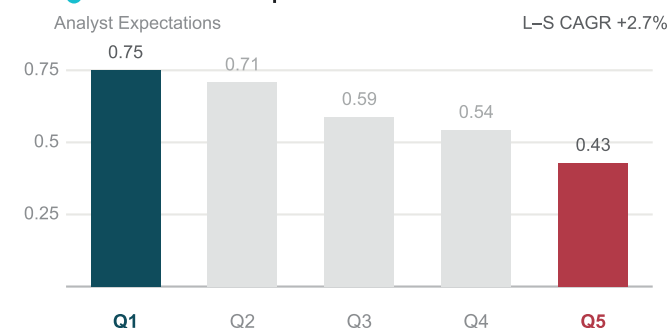
226 ● Net Current Assets to Price Ratio



227 ● Pretax Income to Net Operating Assets



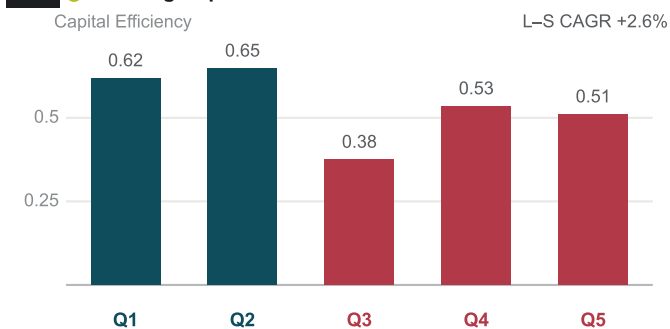
228 ● EBIT Estimate Dispersion



229 ● 1Y Chg in EPS to Operating Cash Flow



230 ● Working Capital Turnover Ratio



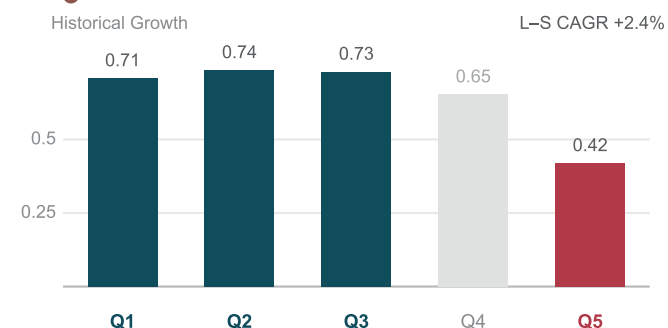
231 ● Change of Sales to Change of EPS



232 ● Pretax Profit Margin



233 ● Sales Acceleration



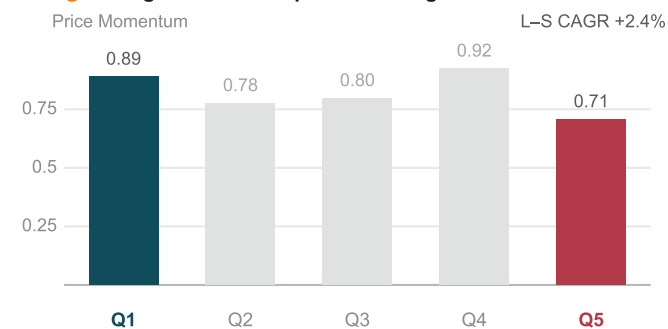
234 ● Risk Adjusted Relative Strength



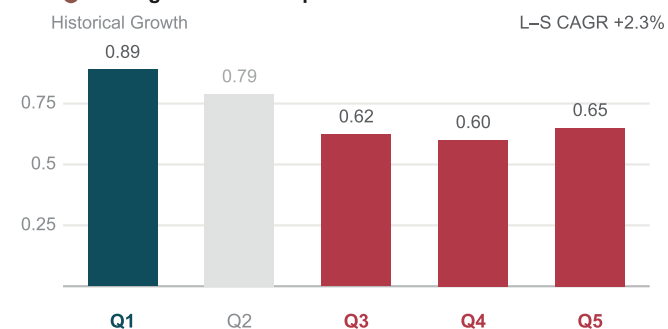
235 ● Analyst Earnings Estimate Diffusion



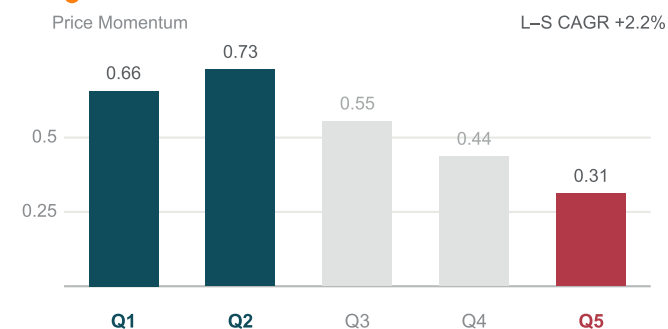
236 ● Change of Ownership Level - Hedge Fund



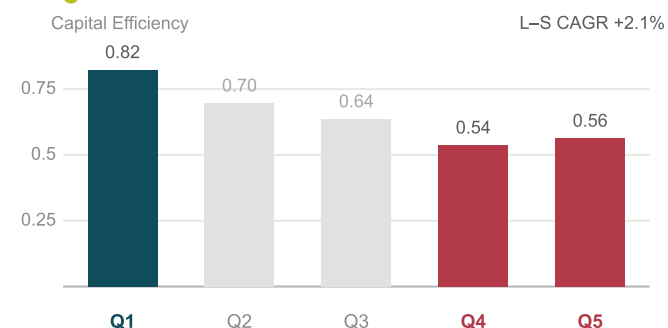
237 ● 3Y Chg in Cash Flow per Share



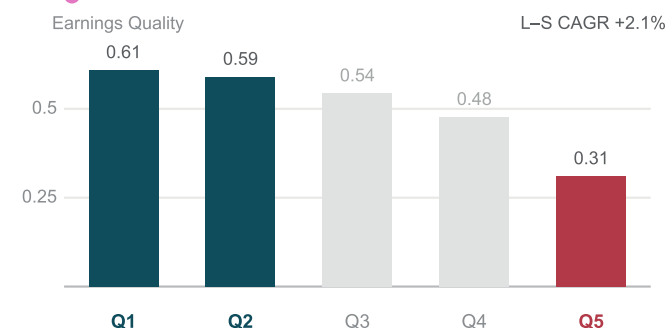
238 ● Investment Duration



239 ● Normalized ROE



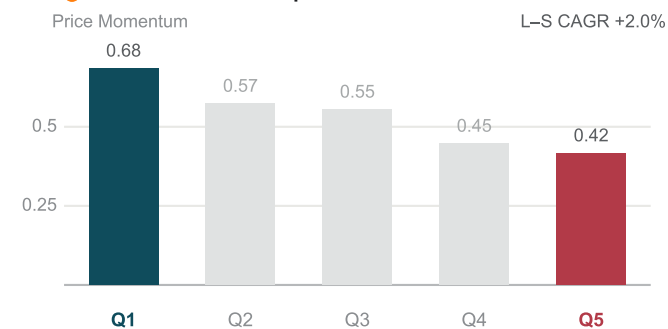
240 ● Cash to Sales



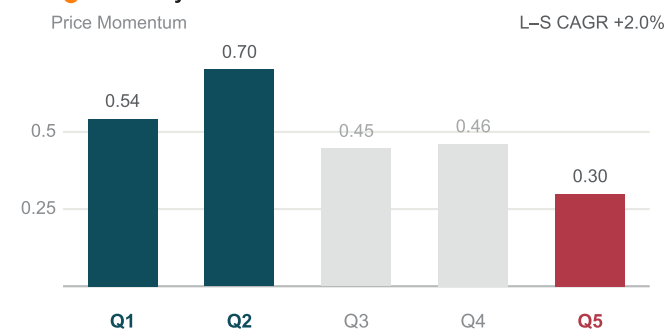
241 ● 3Y Chg in Earnings per Share



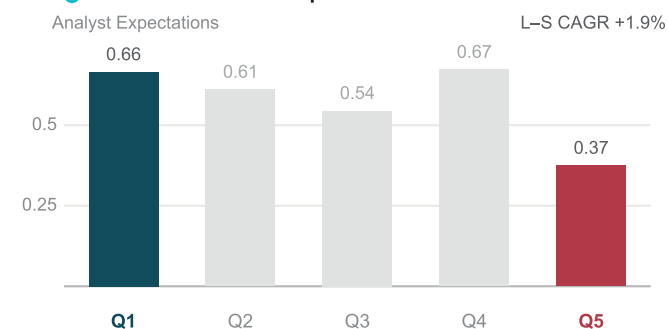
242 ● Institution Ownership Turnover



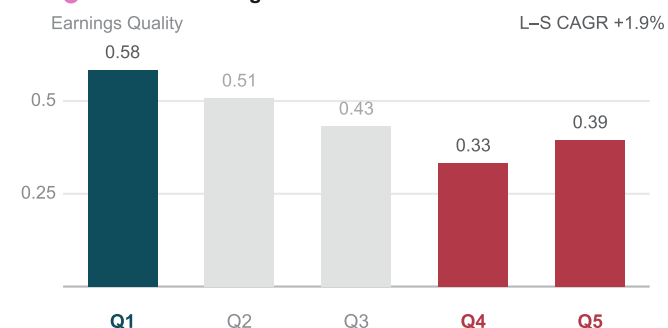
243 ● Max Daily Return in the Past 6 Months



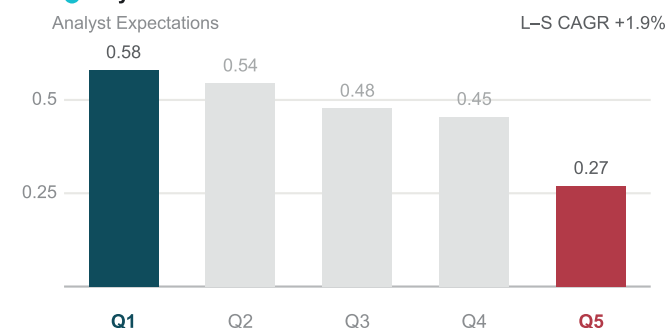
244 ● Revenue Estimate Dispersion



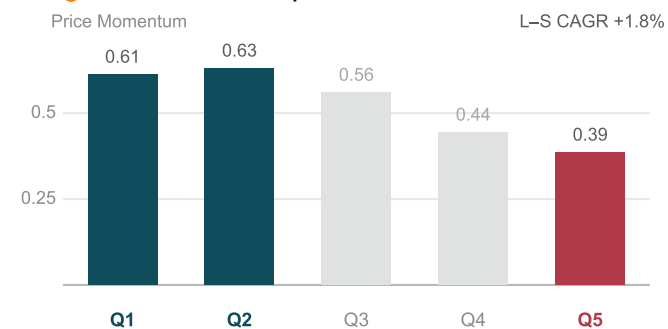
245 ● Retained Earnings to Total Assets



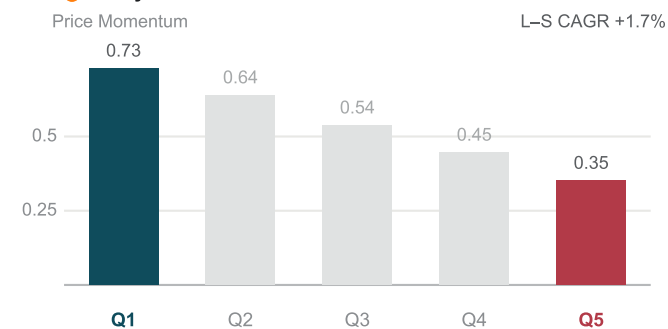
246 ● Buy-to-Sell Recommendation Ratio Less 3M SMA



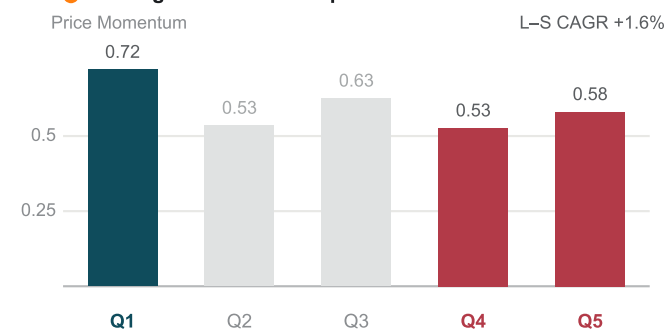
247 ● Institution Ownership Concentration



248 ● 5 Day Price Reversal



249 ● 6M Chg in 18M CAPM Alpha



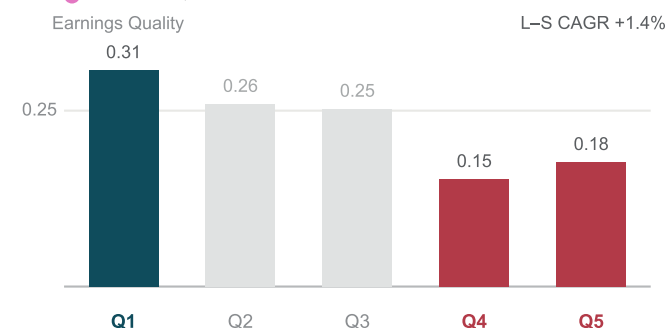
250 ● Year over Year Change of Inventory to Assets



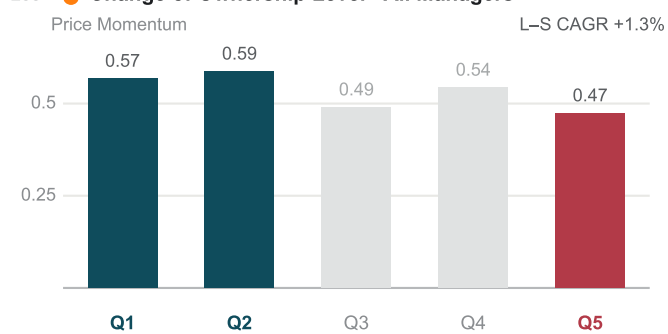
251 ● 6M Chg in 12M CAPM Alpha



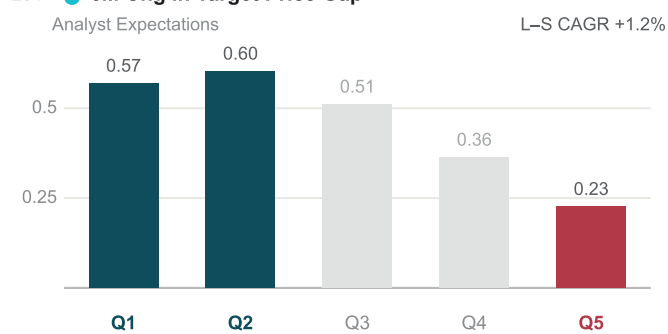
252 ● ROE 20 Qtr Standard Deviation



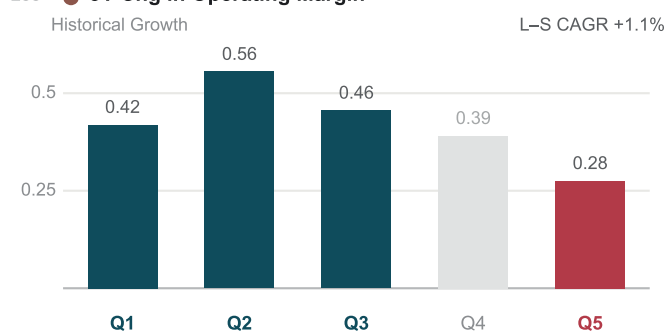
253 ● Change of Ownership Level - All Managers



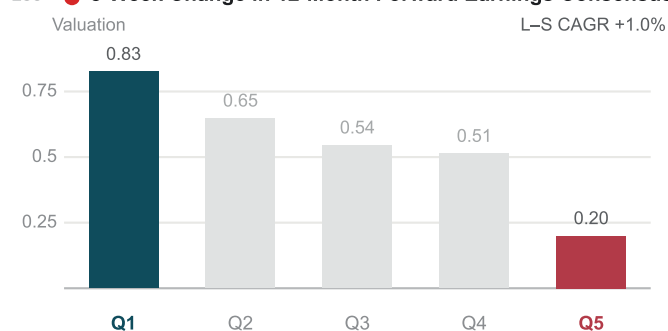
254 ● 6M Chg in Target Price Gap



255 ● 3Y Chg in Operating Margin



256 ● 8-Week Change in 12-Month Forward Earnings Consensus Estima...



257 ● Change in TTM Depr. to Capex



258 ● 6M Avg Chg 1M Recommendation

